

Attack–Fault Identifiability and the Cost of Ambiguity at the Electronic Trading Gateway

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Abstract

Operators of electronic trading gateways must often decide how to respond to suspicious order flow before knowing whether its cause is a cyber compromise, a software fault, or an ordinary market shock. We derive, for a linear-Gaussian model of gateway-observable order flow, an exact criterion — equality of severity-weighted projected offsets — under which a compromise and a fault are observationally indistinguishable, together with a matching cost floor: under this condition, no policy using the observation alone can beat the best constant action, imposing a strictly positive ambiguity cost relative to an oracle. We show this confounding is itself fragile under state-level (rather than output-level) dynamics, requiring the specific pair difference to lie in the classical unobservable subspace of the system — a condition that is trivial for a generic dynamics matrix, though we do not claim real, structured market dynamics are necessarily typical of that generic set. We then report a controlled empirical study, on the ABIDES agent-based market simulator, of how incident-construction, feature access, and calibration choices affect classification and decision cost — explicitly not a test of the theorem above, since none of the tested nonlinear constructions were confirmed to satisfy its exact premise. Across five profiles (four independently-designed injection constructions, one feature-engineering variant), classification and cost outcomes vary substantially with these choices; a full 50-cell cost audit — using a direct class-conditional cost-weighting method rather than single-draw resampling, per external review — finds the fitted classifier beats a naive always-contain policy in all 50 configurations and beats the best constant-action policy in 41 of 50; the 9 exceptions are all low-prior cells ($P(K) \leq 0.20$) in which always-leave dominates, concentrated in one profile (Study 5). We report this as a retrospective, whole-episode diagnostic study with a hypothetical decision-cost evaluation attached, establishing sensitivity to specific, named implementation choices — not a general claim about

linear-to-nonlinear transfer difficulty, not a demonstrated real-time containment benefit (the classifier uses complete-episode features, so no claim is made about value delivered before an incident’s consequences finish accumulating), and not a claim to measured economic harm, since the underlying cost figures price a market-impact model against measured order volume, not independently validated incremental loss. Methodological corrections made during this project’s development, including a prior-shift threshold error that had produced a spurious cost finding, are documented in a supplementary changelog rather than the main text. A practitioner checklist and full per-cell evidence tables, including exact provenance for every headline result, are provided as supplementary material.

Keywords: cyber-physical security, electronic trading, identifiability, decision under uncertainty, incident response

1. Introduction

A trading participant can generate harmful or suspicious orders because of a compromised credential, a software defect, its own order-generation policy, ordinary market conditions, or some combination of these. An operator — a broker or trading firm’s incident-response controller — must often act before the cause is known: throttle, cancel orders, revoke access, or isolate a process. Each of these actions has a cost if applied to a case that did not need it, and each carries risk if withheld from a case that did. **A scope note, stated here rather than deferred to Section 3.5, per external review: this motivating scenario describes a real-time decision problem, but the theoretical and empirical results in this paper concern retrospective diagnosis from complete-episode observations, with a hypothetical decision-cost evaluation attached to the resulting classifications.** We use the real-time motivation to explain *why* the underlying identifiability question matters operationally, not as a claim that this paper’s specific classifiers or cost figures measure the value of a timely intervention — Section 3.5 states the gap between these two framings precisely, and it should be read as qualifying every empirical claim in Sections 5–7, not as an isolated limitation confined to one paragraph.

Two questions precede any policy question. First: **can the causes even be told apart** from the evidence an operator is ordinarily permitted to observe (order acknowledgments, market data, session and process telemetry)? Second, if not: **how costly is that ambiguity**, given that operators must still act on it, typically via a fixed, threshold-based response? Industry practice for algorithmic-trading kill switches and circuit breakers is overwhelmingly of the second kind: trip a threshold (order rate, cancel ratio, loss limit) and act, with an explicit design philosophy of *preserve capital first, diagnose later* — no attempt to establish cause before acting.

This paper answers the first question with a formal identifiability result (Sections 3–4), scoped deliberately to a linear-Gaussian order-flow model — mirroring the approach in the closest structural analog to our problem, the literature on distinguishing cyberattacks from physical faults in cyber-physical systems, which likewise proves distinguishability results only for linearized or additive-unknown-input models. **Section 4.1 then shows this confounding is fragile even within the linear world itself**, once a more realistic state-level injection semantics is used, via an exact connection to classical control-theoretic observability. Sections 5–6 validate the core theorem at pilot scale within that same linear world, including an independent second implementation. **Section 7 is the paper’s central empirical contribution**: a controlled study of how incident-construction choices (which observable statistics are matched between two mechanisms, which features an observer has access to, and how a fitted classifier’s decision threshold is calibrated) affect classification and decision cost in a realistic agent-based market simulator. This is explicitly not a test of whether the theorem above holds in a nonlinear market — the constructions studied were never shown to satisfy the theorem’s premise (equality of the full observation-trajectory law), so no result in Section 7 confirms, contradicts, or measures a “failure to transfer” of Theorem 1. What Section 7 does establish is that matching incident mechanisms on a small number of chosen summary statistics, in a realistic multi-agent market, leaves a highly construction-dependent residual of exploitable information in other channels — a finding about observation-matching practice, not about the linear theorem’s validity.

Contributions. 1. An exact fixed-instance equality criterion for observational indistinguishability between two incident mechanisms in a linear-Gaussian order-flow model ($\theta_K C b_K = \theta_F C b_F$), together with a separate rank-based characterization of which severity/direction pairs can be made to satisfy it (Theorem 1 and Corollary 1 — corrected here from an earlier version

of this list, which called the whole result a single “rank-based necessary-and-sufficient condition,” conflating the fixed-instance equality with the separate existence question), together with a cost-of-ambiguity lower bound (Proposition 1). 2. A pilot-scale simulation validating both predictions within the linear model, plus an independently implemented second simulator (different code structure, dynamics, features, and classifier family) confirming the qualitative pattern is not a code artifact. 3. **A second, exact theoretical result (Theorem 2, Section 4.1) showing Theorem 1’s confounding is itself fragile even without leaving the linear model:** once the incident is modeled as a state-level rather than output-level disturbance — arguably the more realistic choice — confounding requires the far more restrictive condition that the specific pair difference Δ_0 lie in the classical (Kalman) unobservable subspace of the dynamics, which is trivial for a generic (A, C) (though “generic” is a statement about a full-measure set of matrices, not a claim that real, structured market dynamics are typical of that set — Section 4.1 states this qualification explicitly), confirmed both analytically and by a direct empirical demonstration that an exactly-confounded pair becomes highly classifiable within a few time steps under generic coupled dynamics. 4. **A controlled empirical study on ABIDES of how incident-construction choices affect classification and decision cost,** escalated into a systematic sensitivity analysis across four independently-designed injection layers: the accuracy ranking between a summary-matched and a contrast construction is itself highly sensitive to which features an observer has access to, which specific statistic was matched during calibration, classifier capacity, and (per Section 6’s correction) proper threshold calibration under prior shift — with five qualitatively distinct classification profiles observed across four designs. **This is a study of observation-matching and decision-cost behavior, not a test of Theorem 1:** none of the tested constructions were shown to instantiate the theorem’s premise (equality of the full observation-trajectory law), so no claim here concerns the theorem’s validity, only the practical difficulty of engineering a nonlinear analog of exact confounding and the decision-cost consequences of not fully succeeding at it. 5. **A duration/persistence ablation (Section 7.8),** reframed after external review as not causally isolating the mechanism it was designed to test: removing incident persistence collapsed classification accuracy substantially in the tested scenario, directionally consistent with Theorem 2’s state-feedback prediction, but the design conflates persistence with incident duration and cumulative activity, and a follow-up isolation re-

design (Section 7.10.2) remains inconclusive at the sample size tested. 6. A revised, narrower statement of what results of this form (ours and the cyber-physical-systems analog it is built on) can and cannot be assumed to say about real deployed systems, plus open questions — stated as such, not as a settled counter-recommendation (Section 7.11) — about the premises of the companion paper’s (Paper 2) recommendation to target verification investment at the confounding boundary.

2. Related work

Decision-theoretic cyber response. Belief-state (POMDP) formulations of cyber defense track an attacker’s progression through a network and choose interference actions under noisy alerts (Miehling, Rasouli & Teneketzis, 2018); related work formulates response as a Stackelberg game over attack-response trees (Zonouz et al.’s RRE) or balances service continuity against intrusion risk under partial observability (Hammar & Stadler, 2022; Booker & Musman, 2020). None of this literature prices market liquidity or service interruption to legitimate trading. **We correct an overclaim here, per external review:** our own cost structure is a fixed-loss-table/market-impact-proxy analysis (Sections 3.4, 6, 6.3), not a quantified model of endogenous liquidity replacement by other market participants — an earlier version of this sentence claimed the latter was “central to our cost structure,” which overstates what the implemented analysis actually does. The defensible claim is narrower: this literature does not price market-impact costs of the kind Section 6 constructs, not that it omits a component this paper has itself quantified.

Cyber-physical fault-versus-attack differentiation. The closest structural analog to our identifiability question is the literature distinguishing physical faults from cyberattacks in power systems and other cyber-physical systems. Pasqualetti, Dörfler & Bullo (2013) is a foundational precedent here, arguably closer to our Theorem 2 than any other cited work: they develop a geometric framework using the unobservable subspace of a linear system to characterize exactly which attacks are detectable and identifiable by an output-based monitor — the same mathematical object our Theorem 2 uses to characterize when compromise-versus-fault confounding survives state feedback. Roy & Dey (2024) is a more direct methodological

precedent for our Theorem 1 specifically: they model an anomalous input to a linear system, estimate it via a sliding-mode unknown-input observer, and derive a closed-form distinguishability metric with a criterion for when the estimated input is provably a fault versus an attack, including a “fault-mimicking cyberattack” corollary — an algebraic condition under which an attack direction lies in the fault subspace, the direct analog of our Corollary 1. Anwar, Mahmood & Shah (2015) complements this from the data-driven side, reporting that standard supervised classifiers “struggle to distinguish cyber-attacks from physical faults” on realistic smart-grid data. **We note this as a suggestive parallel, not an explained instance of our own result:** their setting (smart-grid state estimation) is a different domain from ours, and we have not established that it satisfies our model’s confounding condition or any analogous one — Proposition 1 provides a mechanism (a provable information-theoretic floor) by which such empirical struggles *can* arise in a system that does satisfy the relevant confounding condition, not a demonstrated explanation of their specific dataset’s classifier failures, which would require checking their system against that condition directly.

Engaging this literature’s methodology directly, not just its conclusions: Roy & Dey’s validation, like our own linear-model pilot (Section 5), is simulation-based rather than tested against a physically realized, nonlinear testbed — their “simulation case studies” instantiate the same linear model their theorem is proved for, so their framework’s internal consistency is well-established, but whether an analogous criterion could be validly instantiated in a genuinely nonlinear physical system (analogous to our ABIDES study) is, to our knowledge, not attempted in their work or elsewhere in this literature. Our Section 7 takes up this open question empirically and reports that constructing a valid nonlinear analog of exact confounding — one that actually matches the full observation law, not merely a chosen summary statistic — is itself a nontrivial, easy-to-get-wrong engineering task, with the resulting classification ranking sensitive to feature access, calibration target, and classifier capacity in ways the linear model does not anticipate. We do not claim to have established whether their criterion’s qualitative implications hold or fail once genuinely instantiated in a nonlinear system, since none of our attempted constructions were confirmed to be valid instantiations; what we do show is that attempting the instantiation is harder, and more sensitive to implementation choices, than the linear theory alone would suggest. We read our contribution, in this literature’s terms, as raising and empirically probing — without fully resolving — a validation question their

own framework does not itself address: what does it take to correctly realize a linear-derived distinguishability criterion’s premise once observations come from genuinely nonlinear, compounding dynamics rather than being assumed linear from the outset?

A separate, more recent line (Alisic & Amin, 2026) approaches a related transfer question from the opposite direction — geometrically, via Lie groups, rather than empirically via simulation — proving a precise sufficient condition (Lie-bracket commutativity) under which sensor-attack stealthiness transfers from linear to nonlinear operating conditions. Their result and ours are related but not directly comparable: theirs is a proved sufficient condition for a specific transfer property in a specific technical setting; ours is an empirical study of how difficult it is to engineer a nonlinear analog of exact confounding in a realistic market, without deriving or checking an analogous condition for our setting. **We do not claim, and withdraw any earlier suggestion, that our market “fails” their condition or “whatever the analogous condition would be”** — no such condition has been derived for the order-flow/observation-matrix structure of Section 3, so no claim about satisfying or failing it is supported. The genuine open question, stated at the correct strength, is whether a Lie-group-style commutativity condition *could* be formulated for our setting, and if so, whether it would help explain the construction-dependence Section 7 documents — a question this paper raises but has not answered.

Market manipulation detection and trading kill switches. A large literature detects manipulative patterns (spoofing, wash trading) from trade and order-book data, and a 2025 conceptual paper explicitly titled around “cyber risk containment in algorithmic trading” surveys AI-driven detection, blockchain integrity, and zero-trust architectures for trading infrastructure, but offers no decision-theoretic formalism, no identifiability analysis, and no treatment of liquidity as an endogenous cost. **A precision worth stating explicitly, per external review: real-world kill-switch implementations and patents describe threshold-based anomaly triggers (order-flow covariance inflection points, cancel-ratio thresholds) — these are *observation-dependent* policies, responding to data, not the *observation-independent constant* policy (“always contain”) this paper uses as one baseline.** Citing threshold-based industry practice does not, by itself, validate a constant action as a realistic operational comparison — the two are related (both act without first diagnosing cause) but are not the same object, and a reader wanting an operational compari-

son would need a specified threshold detector with comparable feature access and timing, which this paper does not provide. The constant-action baselines in this paper should be read as decision-theoretic reference points (the best achievable without any observation at all), not as a claim that real kill-switches are literally constant policies. Separately, empirical work on automating stock-market manipulation via compromised brokerage accounts confirms that compromise and manipulative order-generation behavior are a realistic joint failure mode, supporting our model’s treatment of compromise as a distinct, structured perturbation to order flow rather than an unstructured anomaly.

Costly information acquisition. A separate literature treats costly, endogenous information acquisition in asset markets (e.g., models in the Grossman–Stiglitz tradition, and recent optimal-stopping formulations of costly sequential information acquisition). This addresses *traders’* incentives to learn about asset value, not an *operator’s* decision to verify an incident’s cause before containing it, and does not engage with observational indistinguishability at all; we treat it as a technique-adjacent rather than a competing literature.

Scope of the novelty claim, stated precisely rather than sweepingly. We do not claim no existing work touches any combination of identifiability analysis, market cost, and fixed-policy critique — the CPS distinguishability literature (Roy & Dey; Anwar, Mahmood & Shah) and the Lie-group transfer literature (Alisic & Amin) share substantial theoretical machinery with Sections 3–4, and the market-manipulation-detection and kill-switch literatures independently establish parts of our motivating premise. What we have not found is the specific combination this paper assembles: a corrected, formally exact identifiability and cost-floor result for the compromise-versus-fault question *in the electronic-trading setting specifically*, together with a controlled empirical study of how incident-construction choices affect classification and decision cost in a realistic nonlinear market simulator, and a corrected decision-cost benchmark distinguishing the best-achievable-using-observations risk from the best-constant-action risk — a distinction an intermediate version of this paper itself initially conflated, and whose correction is itself part of the contribution. This is a narrower and more defensible claim than an unqualified “no existing work combines...”, which is withdrawn.

An explicit, itemized incremental-contribution comparison against the two closest predecessors, added per external review’s finding that mathematical correctness alone does not establish a suffi-

ciently new contribution. Roy & Dey (2024) derive a fault-mimicking distinguishability criterion via a sliding-mode unknown-input observer and validate it in linear simulation; Pasqualetti, Dörfler & Bullo (2013) establish the classical unobservable-subspace machinery for attack detectability and identifiability that Theorem 2 specializes. Neither of these — nor, to our knowledge, any other work in this literature — does the following, which is what this paper actually adds: (1) applies this style of criterion to the specific loss structure of a trading gateway (compromise vs. fault vs. an oracle, with a market-impact-based cost floor, rather than a generic detection-accuracy or false-alarm-rate metric); (2) derives the state-feedback fragility result of Theorem 2 as a *further* condition beyond the one-shot output-level criterion both prior works implicitly assume is stable under feedback, which we show it is not, generically; (3) empirically tests — rather than assumes — whether a linear-model identifiability criterion of this type transfers to a genuinely nonlinear, multi-agent instantiation, finding that constructing a valid nonlinear analog is itself a nontrivial, easy-to-get-wrong exercise across four independent attempts; and (4) documents, as a methodological contribution in its own right, a specific and generalizable failure mode (prior-shift threshold miscalibration manufacturing a spurious cost-sensitivity finding) that is not specific to this paper’s domain and could recur in any cost-sensitive decision pipeline built on a classifier trained at one class balance and deployed at another. **We consider (3) and (4) the paper’s most defensible claims to novelty; (1) and (2) are closer to careful, correctly-scoped applications of established machinery than new theoretical results, and we do not claim otherwise.** Whether this combination clears the bar for the ambitious venues named in Section 10’s target-venue discussion is a judgment we leave to reviewers at those venues, not one we assert here.

3. Model

Domain grounding for the K/F labels, added per external review: these labels name programmed order-generation mechanisms in the formal model and the ABIDES simulation, not a general claim about compromise or fault in the wild. Throughout this paper, K (“compromise”) and F (“fault”) denote specific, fully-specified perturbations to the order-flow process (Sections 3.2 and 7’s ABIDES designs) — a

directional bias, an elevated order rate, or an altered cancellation pattern, applied under a chosen severity and onset time. We do not claim these mechanisms are representative of any actual attacker’s access, objective, or capability, nor of any specific real software defect; an attacker could in principle imitate a fault’s statistical signature, and a fault could in principle produce strongly directional order flow, either of which would blur the distinction this paper’s labels assume is clean. What this paper’s identifiability and classification results establish is *behavioral-mechanism identification* — whether two specified generative processes are distinguishable from gateway-observable order flow — not *inference about intent or authorization*, which would require a different kind of evidence entirely (e.g., credential logs, code-change records) that this paper does not model. The four ABIDES designs used in Section 7 are best understood as four independently-*parameterized* scenario variants sharing common simulation infrastructure, not four independent software implementations, nor a representative sample of the space of possible real incidents (Section 9’s independence-classification item states this explicitly).

3.1. Order-flow dynamics

Let $x_t \in \mathbb{R}^n$ denote a latent representation of normal order-flow state (e.g., components loading onto price pressure, volume intensity, and order-cancellation propensity), evolving under known linear dynamics:

$$x_{t+1} = Ax_t + w_t, \quad w_t \sim \mathcal{N}(0, Q).$$

Assumptions made explicit, per external review, rather than left implicit: $\{w_t\}$ and $\{\varepsilon_t\}$ (Section 3.3) are independent draws across time and across each other in this project’s implementation — independence *is sufficient* to construct the joint Gaussian trajectory Theorem 1’s “Gaussian trajectory” language requires, but it is a *stronger* condition than joint Gaussianity in general (jointly Gaussian variables need not be independent), so we correct an earlier, imprecise version of this sentence that called independence “not an additional assumption beyond” joint Gaussianity — it is a specific, sufficient way of satisfying that requirement, not an equivalent restatement of it. The initial state x_0 is implemented as a deterministic point ($x_0 = 0$), not drawn from a nondegenerate distribution — a degenerate special case of “Gaussian” (a point mass has zero covariance), stated precisely here rather than implying a general Gaussian prior over initial states. The finite-horizon

experiments in Sections 5–6 always include timesteps at and after the incident onset τ by construction ($T = 40$, $\tau = 20$, so $T > \tau$ unconditionally in every run) — Theorem 1’s distinguishability claim would be vacuous on a window ending before τ , and this project’s implementation never uses such a window.

3.2. Incident injection

At a (possibly unknown) onset time τ , an incident of mechanism $M \in \{K, F\}$ (compromise or fault) injects an additive perturbation into the pre-observation order-flow signal:

$$\tilde{x}_t = x_t + \mathbb{I}[t \geq \tau] \theta_M b_M, \quad b_M \in \mathbb{R}^n, \theta_M \in \mathbb{R}.$$

b_M is a mechanism-specific *injection direction* (how a compromise versus a fault characteristically distorts order flow); θ_M is a scalar severity/magnitude.

3.3. Gateway-observable output

The operator observes only a lower-dimensional, noisy projection of order flow through a fixed, known observation matrix $C \in \mathbb{R}^{m \times n}$ ($m < n$), representing what a gateway or market-data feed can actually surface (e.g., spread, depth, order-cancellation rate):

$$y_t = C \tilde{x}_t + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \Sigma).$$

Everything except the injection term (A , Q , C , Σ , and the pre-incident distribution of x_t) is common across hypotheses about M .

3.4. The loss table, and what it assumes (added per external review)

The cost analysis in Sections 6–7 uses the two-action, two-cause loss table

Action	Compromise K	Fault F
Contain	C_{OP}	DISRUPT
Leave alone	HARM	0

This table encodes a specific scenario, not a general property of compromises and faults, and we state that scenario explicitly rather than leaving it implicit: it assumes there exists a specific class of fault for which containment is unnecessary and leaving the system running

incurs zero *incremental* loss relative to doing nothing — for example, a fault that produces anomalous-looking but ultimately harmless order-flow noise (a logging error inflating apparent order counts, or a benign retry loop that does not cross the spread), as opposed to a fault that itself generates destructive orders requiring intervention regardless of cause. The paper’s motivating discussion elsewhere also mentions harmful software defects that plausibly *would* require containment — for that broader class of fault, the correct loss table has DISRUPT replaced by whatever cost containing a harmful-but-benign-cause incident actually carries, and the fault’s own leave-alone cost would not generally be zero. **The strict value of distinguishing the two causes is zero whenever some single action is simultaneously optimal for both** — Proposition 1’s cost floor is a property of *this* loss structure (and, more generally, of any loss structure where the causes call for genuinely different optimal actions), not evidence that ambiguity about cause is always costly regardless of what the two causes actually require.

This assumption was audited directly against the actual simulated fault mechanisms, per external review, and largely does not hold for the ABIDES results in Section 7 — a consequential finding, stated plainly rather than softened. Every one of the four ABIDES designs’ fault (F) mechanisms submits real limit orders unconditionally (`place_incident_order()` calls `place_limit_order()` for K , F , and baseline alike, confirmed by direct source inspection) — there is no “harmless, logging-only” fault anywhere in this project’s actual implementation, only the illustrative logging-error/retry-loop example given above, which does not describe any simulated mechanism. Auditing each design’s specific F parameterization against whether it is non-directional, non-elevated in size/rate, and free of price aggressiveness (the properties that would make “leave alone” plausibly costless): **an explicit, per-design classification, replacing an earlier and inconsistent “three of four” summary count** — V1 confounded: fails (directionally biased and volume/price-elevated); V2 confounded: fails (same reasons); V3 confounded: questionable (non-directional, but meaningfully elevated order volume); V4 confounded: questionable (same reasons as V3). **Only the contrast regime’s F mechanism in V2 and V3 specifically (non-directional, no size/rate/price elevation, only added cancellation churn) plausibly satisfies the zero-incremental-loss assumption as stated** — V1’s and V4’s contrast-regime F mechanisms are also volume-elevated and therefore also questionable, not clean. In short: 0 of 8 audited cells (4 designs $\times$ 2 regimes) are clean passes

without qualification; 2 of 8 (V2 and V3’s contrast regimes) are plausible passes; the remaining 6 range from questionable to a clear failure of the assumption.

What this means for Section 7’s dollar-cost tables: the illustrative cost reconstruction in Sections 6.2 and 7.7 is defensible as a hypothetical decision-cost scenario — a chosen loss table applied consistently to observed classification outcomes — but should not be read as an economically accurate costing of these specific simulated incidents, since most of them do not satisfy the scenario’s own defining assumption (that leaving F alone is actually free), and, per external review, since **the underlying quantity itself is a modeled execution-cost proxy applied to submitted order volume, not a directly measured economic loss** — submitted volume need not equal executed volume (cancellations and partial fills intervene), and positive execution costs are not the same claim as positive *incremental* harm caused by F specifically, since ordinary baseline trading also has positive execution costs. Readers should treat Section 7’s dollar figures as “what this specific illustrative loss table implies, applying a market-impact cost model to measured submitted volume,” not as validated economic costs of the actual simulated fault behavior or its incremental harm relative to a baseline. **We have since attempted the corrected loss table this section originally deferred (Section 6.3) — after two real costing bugs in our own formula-based attempts were found and fixed, a final costing from directly-measured submitted order volumes (not executed volume, and not independently validated as incremental harm) shows F ’s order flow is nonzero-cost under this model, contrary to the original loss table’s zero assumption, at a scale (\$0.85K–\$104.67K depending on design and regime) far below the illustrative cost triple’s other terms.** We withdraw an earlier, overstated version of this sentence that claimed this “confirms... this simplification does not change which policy is preferred in any of the 40 tested configurations” — the full 50-cell audit (Section 7.7) shows this is not uniformly true (9 of 50 cells, all at $P(K) \leq 0.20$ and 6 of them Study 5, fail to beat the best constant policy, though every cell beats always-contain). What can be said at the level of evidence actually available: across the five profiles’ mean-across-priors summaries, the qualitative pattern (four profiles’ classifiers substantially beat the best constant policy; Study 5’s does not at low priors) is the finding this correction actually supports.

3.5. Observation timing and scope (added per external review)

The motivating framing in Section 1 emphasizes acting *before* a full diagnosis is possible — an urgent-response setting. **The formal model and pilot experiments in this paper do not fully match that framing, and we state the mismatch explicitly rather than leave it implicit.** Theorem 1 and Proposition 1 concern the full infinite (or fixed-horizon) observation trajectory $\{y_t\}_{t \geq 0}$; Section 5’s pilot uses whole-episode summary statistics (mean, standard deviation, net drift over the full 40-step window) as features, and Section 7’s ABIDES experiments likewise use full-episode summaries rather than a prefix available at some specific, pre-committed decision deadline. **This is, as implemented, closer to an offline or retrospective diagnosis study than to a real-time response-deadline study:** nothing in the current pipeline enforces that features are computed only from information available at the moment an operator would actually have to act, and the classifier is not “charged” for including post-decision information the way a genuinely online policy would need to be. A paper making a strict real-time-response claim would need to define an explicit observation cutoff and decision deadline, and evaluate classifiers using only prefix features available at that cutoff — we have not done this, and the results in Sections 5–7 should be read as characterizing what is discoverable given the full episode, not what is discoverable in time to inform the specific urgent action Section 1 motivates. This does not invalidate the identifiability results (which are about what a trajectory’s full distribution reveals, a well-posed question independent of timing), but it does mean the cost figures in Sections 6–7 should not be read as measuring the value of information available at a real operational decision point without this additional restriction.

4. Theoretical results

Theorem 1 (Indistinguishability criterion). Under the model of Section 3, with common onset time τ , mechanism instances (K, θ_K, b_K) and (F, θ_F, b_F) induce identical distributions over the observation trajectory $\{y_t\}_{t \geq 0}$ if and only if

$$\theta_K C b_K = \theta_F C b_F. \quad (*)$$

Proof. The trajectory $\{y_t\}$ is Gaussian with a mean path determined entirely by $\{C\mathbb{E}[\tilde{x}_t]\}$ and a covariance path determined by A, Q, Σ , none of which depend on M . For $t < \tau$, the mean and covariance are identical under both hypotheses by construction. For $t \geq \tau$, the mean path differs only by the constant offset $C\theta_M b_M$ added to the shared mean $C\mathbb{E}[x_t]$ (a mean, not the random vector Cx_t itself — corrected per external review); the covariance is unaffected by M . Two Gaussian processes with identical covariance structure and mean paths differing only by a constant offset are equal in distribution if and only if that offset is equal, i.e. iff $(*)$ holds. ■

Corollary 1 (Fault-mimicking compromise — corrected for degenerate cases). Fix $u = Cb_K$, $v = Cb_F$, and a nonzero compromise severity $\theta_K \neq 0$. If the fault severity θ_F may be any real number, then

$$\exists \theta_F \in \mathbb{R} : \theta_K u = \theta_F v \iff u \in \text{span}(v).$$

This is the correct statement; the original phrasing (“rank = 1”) was imprecise in two ways, now fixed: (i) $\theta_K = \theta_F = 0$ trivially satisfies $(*)$ for *any* b_K, b_F and must be excluded by fixing $\theta_K \neq 0$ as above, not left implicit; (ii) linear dependence corresponds to $\text{rank} \leq 1$, not $\text{rank} = 1$ exactly — both u and v could be zero, or one could be zero while the other is not, and these subcases behave differently (if $v = 0$ but $u \neq 0$, no θ_F satisfies the equation; if both are zero, every θ_F trivially does). If both severities are additionally required to be nonzero, the condition sharpens to: either $u = v = 0$, or u, v are both nonzero and collinear. If severities must be positive, nonzero u, v must lie on the same ray, not merely the same line. For a *family* of admissible fault directions $\{b_F^{(i)}\}$, we distinguish two distinct claims the original text conflated: **choosing** one fault direction gives fault-mimicking whenever u lies on the line through some single $Cb_F^{(i)}$ (a union of lines); **superposing** multiple fault directions (if the generative model allows combined faults) gives the weaker requirement $u \in \text{span}\{Cb_F^{(i)}\}$ (a full subspace or cone). The manuscript’s claims use the span form and should be read as assuming fault superposition is a modeled possibility, not merely a choice among single fault directions — this should be stated explicitly wherever Corollary 1 is invoked.

We also flag, rather than resolve, a scope question raised by this correction: establishing that *some* instance (F, θ_F, b_F) matches *some* instance (K, θ_K, b_K) is a statement about overlapping parameter families, not a statement that $P(Y \mid M = K) = P(Y \mid M = F)$ after marginalizing over a distribution on severities, onset times, or directions within each mechanism

class. Different fixed severities on the same projected direction produce different means; a composite/marginal indistinguishability claim requires either a pairwise-instance reading (which is what Theorem 1 and this corollary literally give) or an explicit mixture model over the mechanism class, which this paper does not specify. All results below should be read as pairwise-instance statements unless a mixture model is given.

Corollary 2 (Separation implies distinguishability, not high accuracy). If $\text{rank}\begin{pmatrix} Cb_K & Cb_F \end{pmatrix} = 2$, no nonzero θ_K, θ_F satisfy (*); for any instance pair with $(\theta_K, \theta_F) \neq (0, 0)$, K and F are then distinguishable in the population limit. This does **not** by itself imply near-perfect or even moderately good finite-sample classification — the projected directions can be arbitrarily small relative to noise. The correct finite-sample benchmark uses the *whitened* mean separation, not either mean offset scaled by noise magnitude alone: for $Y \mid M \sim \mathcal{N}(\mu_M, V)$ with $V \succ 0$, define $\delta = \mu_K - \mu_F$ and the (squared) discriminability $d^2 = \delta^\top V^{-1} \delta$. Under equal priors, the Bayes-optimal accuracy and AUC are the standard Gaussian-discrimination identities

$$\text{Acc}^* = \Phi(d/2), \quad \text{AUC}^* = \Phi(d/\sqrt{2}),$$

obtained by projecting onto the log-likelihood-ratio score (mean $\pm d^2/2$, variance d^2 under each hypothesis). These are standard results from classical signal detection theory (see, e.g., Green & Swets, 1966, or any standard detection-and-estimation reference such as Poor, 1994), not a contribution of this paper, but they provide the correct analytical benchmark against which Section 5’s simulated accuracy/AUC values should be compared, rather than treating “rank 2” alone as a qualitative promise of strong separation.

A notational precision worth stating explicitly, per external review: Y in this paper sometimes denotes the raw observation trajectory and sometimes the engineered summary vector supplied to a classifier — these are different objects with different achievable risks. Writing $Z = \phi(Y)$ for a feature map applied to the raw trajectory, the correct ordering is $R_{\text{oracle}} \leq R_Y^* \leq R_Z^* \leq R_{\text{const}}$, with $R_Z^* \leq R(\hat{\pi}(Z))$ for any fitted policy $\hat{\pi}$ — a feature ablation changes Z and can only move R_Z^* , it does not change Y ’s complete observation law or R_Y^* . Section 6’s discussion of the non-Gaussian ABIDES feature summaries should be read as concerning R_Z^* , the best achievable risk given the engineered features actually used, not R_Y^* , the best achievable risk given the full raw trajectory — these coincide only

when ϕ is sufficient for the classification problem, which is not established for any feature set used in this paper.

An unequal-prior extension of the Gaussian benchmark above, supplied per external review, completing the decision-cost analog of Corollary 2’s discriminability formulas. For the loss structure of Section 6 ($a = p(H_K - C_K)$, $b = (1 - p)(C_F - H_F)$) applied to the equal-covariance Gaussian raw-trajectory setting above, direct likelihood-ratio integration gives the population-optimal ambiguity cost as a function of discriminability d :

$$\text{CoA}_Y^* = a \Phi\left(\frac{\log(b/a)}{d} - \frac{d}{2}\right) + b \Phi\left(\frac{\log(a/b)}{d} - \frac{d}{2}\right),$$

which reduces to $\min(a, b)$ at $d = 0$ (exact confounding, recovering Proposition 1) and decreases toward 0 as $d \rightarrow \infty$ (perfect separation). This is a standard consequence of Gaussian hypothesis testing, not a new result, but it supplies the missing unequal-prior cost benchmark for the raw-trajectory case; comparing any of this paper’s summary-based classifiers against it should be done while explicitly acknowledging that a summary statistic is an information-restricted version of the raw trajectory, so $R_Z^* \geq \text{CoA}_Y^* + R_{\text{oracle}}$ in general, not equality.

Proposition 1 (Cost of ambiguity is bounded below under confounding — corrected). *The version of this proposition in an earlier draft of this manuscript was false as stated: the stated assumptions (DISRUPT $\neq 0$, HARM $\neq C_{OP}$) admit a counterexample — $p = 1/2$, $C_{OP} = 3$, DISRUPT = 6, HARM = 1 satisfies both assumptions, yet leaving alone is cost-optimal for both true causes (HARM = 1 < $C_{OP} = 3$), so the ambiguity cost is exactly zero. We are grateful to an external review for identifying this and are correcting it here rather than leaving the earlier version standing. The missing assumption is that harm must strictly exceed the containment cost.*

Suppose (*) holds (the confounded case), $0 < p < 1$, DISRUPT > 0 , and HARM $> C_{OP} \geq 0$ (harm strictly worse than routine containment — the assumption the earlier version omitted). Then the best achievable expected cost among policies mapping $\{y_t\}$ to $\{\text{contain, leave alone}\}$ is

$$R^* = \min\{p C_{OP} + (1 - p) \text{DISRUPT}, \quad p \text{HARM}\},$$

the oracle’s expected cost is $R_{\text{oracle}} = p C_{OP}$, and

$$\boxed{\text{CoA}^* = R^* - R_{\text{oracle}} = \min\{(1-p)\text{DISRUPT}, p(\text{HARM} - C_{OP})\} > 0.}$$

Proof. Under (*), the observations carry no information about the cause, so any randomized policy containing with probability r has expected cost $r[pC_{OP} + (1-p)\text{DISRUPT}] + (1-r)p\text{HARM}$, which is affine in r ; its minimum over $r \in [0, 1]$ is therefore attained at a constant action ($r = 0$ or $r = 1$), giving R^* as stated. The oracle observes the true cause and contains iff $M = K$ (optimal since $C_{OP} < \text{HARM}$) and leaves iff $M = F$ (optimal since $0 < \text{DISRUPT}$), giving $R_{\text{oracle}} = pC_{OP}$. Subtracting and simplifying the min gives the boxed expression, which is strictly positive for $p \in (0, 1)$ under the stated assumptions. ■

For a general finite action/cause loss table $L(a, M)$, the corresponding general expression is $\min_a\{pL(a, K) + (1-p)L(a, F)\} - p\min_a L(a, K) - (1-p)\min_a L(a, F)$, which is strictly positive exactly when no single action is simultaneously optimal for both causes at positive probability — the property Proposition 1’s added assumption ($\text{HARM} > C_{OP}$, together with $\text{DISRUPT} > 0$) is what guarantees in the two-action case.

A more general, non-exact-equality version of this result, proposed by external review rather than originating in this paper’s own derivation, and adopted here with attribution (developed to make the decision analysis meaningful under approximate rather than exact confounding, since exact equality is a useful theoretical control case but a brittle operational target). Let f_K, f_F be the observation densities, $a = p(\text{HARM} - C_{OP})$, $b = (1-p)\text{DISRUPT}$. Pointwise minimization of the density-weighted regret gives

$$\text{CoA}^* = \int \min\{af_K(y), bf_F(y)\} dy = \frac{a + b - \int |af_K - bf_F|}{2},$$

and, writing $\text{TV}(P_K, P_F)$ for total variation distance, this identity connects to the classical relationship between Bayes risk and total variation distance in binary hypothesis testing (see, e.g., Le Cam, 1986, or the Scheffé identity underlying it):

$$\min(a, b)(1 - \text{TV}(P_K, P_F)) \leq \text{CoA}^* \leq \min(a, b),$$

which recovers the exact-confounding Proposition 1 when $\text{TV} = 0$. If a common surrogate law Q satisfies $\text{TV}(P_K, Q) \leq \epsilon_K$ and $\text{TV}(P_F, Q) \leq \epsilon_F$,

the triangle inequality gives $\text{CoA}^* \geq \min(a, b) \max\{0, 1 - \epsilon_K - \epsilon_F\}$ — stating precisely the extra premise any “approximate confounding transfers approximately” argument needs: a controlled bound on distributional approximation error, not merely a match on one summary statistic. This is the correct standard against which Section 7’s ABIDES scenarios should be judged, and one important empirical limit applies: **a classifier performing near chance cannot certify a small total variation distance** — failure of specific tested classifiers is evidence of a ceiling on what *those* methods extracted, not proof of an upper bound on total variation or on what any classifier could extract.

This result is the formal content behind the paper’s title: *ambiguity has a cost that no amount of passive observation, however cleverly analyzed, can eliminate* — a limit, not merely an empirical shortfall of current classifiers.

4.1. Confounding is fragile even within the linear model: a state-feedback extension

Theorem 1’s confounding condition has a hidden assumption worth making explicit: the incident is modeled as an **output-level** (sensor-level) disturbance — it shifts x_t only at the point of observation ($\tilde{x}_t = x_t + \theta_M b_M$), while the state’s own recursion ($x_{t+1} = Ax_t + w_t$) evolves as if the incident never happened. This is a real simplification. A cyber compromise or a persistent order-generation fault plausibly behaves more like a **state-level** (process) disturbance: the perturbed behavior at time t feeds into what the system does at $t + 1$, not just into what is observed at t — exactly how an incident agent’s behavior in a real order book feeds into subsequent book state, which other agents then react to.

We show that this distinction is not cosmetic: replacing the output-level model with $x_{t+1} = A\tilde{x}_t + w_t$ for $t \geq \tau$ (the perturbation propagates) makes Theorem 1’s confounding condition dramatically more restrictive, and generically fails to hold for exactly the (b_K, b_F) pairs Corollary 1 constructs.

Theorem 2 (State-feedback confounding requires unobservability). Under the state-level injection model, let $\Delta_0 = \theta_K b_K - \theta_F b_F$. Mechanism instances (K, θ_K, b_K) and (F, θ_F, b_F) are observationally indistinguishable at every $t \geq \tau$ if and only if Δ_0 lies in the unobservable subspace of the pair (A, C) , i.e.

$$\Delta_0 \in \bigcap_{k=0}^{n-1} \ker(CA^k),$$

the classical (Kalman) unobservable subspace of (A, C) , where n is the state dimension.

Proof sketch. Let e_t denote the mean difference in state trajectories between the two hypotheses (noise cancels in expectation). For $t < \tau$, $e_t = 0$. At $t = \tau$: $\tilde{e}_\tau = \Delta_0$. Propagating, $e_{\tau+1} = A\Delta_0$, so $\tilde{e}_{\tau+1} = A\Delta_0 + \Delta_0 = (I + A)\Delta_0$; inductively $\tilde{e}_{\tau+k} = \left(\sum_{j=0}^k A^j\right) \Delta_0$. The observation-mean difference at $\tau+k$ is $C\tilde{e}_{\tau+k}$. Indistinguishability for all $k \geq 0$ requires this to vanish for every k ; since the $k = 0$ term already forces $C\Delta_0 = 0$, the $k = 1$ term reduces to $CA\Delta_0 = 0$, and inductively $CA^k\Delta_0 = 0$ for every $k = 0, \dots, n-1$ (higher powers are linear combinations of these by Cayley–Hamilton), which is exactly membership in the unobservable subspace. ■

A finite-horizon statement, added per external review to make explicit that this result does not require infinite data to characterize. For H post-onset observations, define the extended observability-style

matrix $G_H = \begin{bmatrix} C \\ C(I + A) \\ \vdots \\ C \sum_{j=0}^{H-1} A^j \end{bmatrix}$. Under the stated persistent deterministic injection

and common-noise model, equality over this finite window is equivalent to $G_H\Delta_0 = 0$, which — again by the same telescoping argument as above — is itself equivalent to $CA^j\Delta_0 = 0$ for $j = 0, \dots, H-1$. For $H \geq n$, Cayley–Hamilton extends this to the full, all-time condition of the Theorem. **The result is therefore not inherently asymptotic:** exact equality over any window of at least n post-onset observations already forces the all-time condition; what remains genuinely dependent on horizon and noise level is *practical, finite-sample discriminability* when the condition fails, which Theorem 2’s population-level statement does not by itself bound (see qualification 3 below).

Corollary 3 (Theorem 1’s confounded pairs are generically state-feedback-fragile). Theorem 1 only requires $\Delta_0 \in \ker(C)$ — a single condition. Theorem 2 requires Δ_0 to survive n such conditions simultaneously ($\ker(C), \ker(CA), \dots, \ker(CA^{n-1})$). Whenever (A, C) is observable (for fixed $C \neq 0$, the generic case for a randomly chosen A), the unobservable subspace is $\{0\}$, so **the only state-feedback-robust confounding is the trivial one where $\theta_K b_K = \theta_F b_F$ exactly as vectors** — i.e., the two mechanisms must push the identical linear combination of latent state variables, not merely produce the identical *projection* under C . Theorem 1’s construc-

tion (Section 5, using C 's null space to make $b_K \neq b_F$ while $Cb_K = Cb_F$) is exactly the kind of pair Corollary 3 says will generically fail to remain confounded once fed back into realistic, coupled dynamics.

Empirical confirmation. Using the exact (A, C, b_K, b_F) from Section 5's pilot, with $A = 0.6I$ (a scalar multiple of the identity), classification stays at chance (accuracy 0.48–0.54 across magnitudes 0.6–2.0) even under state-level feedback. **The relevant fact here is simpler than commutativity in general** (a claim that only makes sense for compatible square matrices, and is not the precise mechanism at work): since $A = 0.6I$, we have $CA^k = 0.6^k C$ for every k , so $\ker(C)$ is trivially A -invariant, making this a non-generic, measure-zero special case where Theorem 1's weaker condition happens to already satisfy Theorem 2's stronger one. Replacing A with a genuinely coupled matrix (off-diagonal terms, spectral radius 0.64, confirmed fully observable with (A, C) via $\text{rank}[C; CA; CA^2] = 3$) and re-running the identical experiment at three injection magnitudes: accuracy under state-level feedback rises to 0.71, 0.89, and 0.97 at magnitudes 0.6, 1.2, and 2.0 respectively — compared to 0.52 for the trivial $A = 0.6I$ case at the same (0.6) magnitude — while the output-level model (unchanged) stays at chance throughout. **The same (b_K, b_F) pair that is exactly, provably confounded under the output-level model becomes almost perfectly classifiable within a handful of time steps once the identical perturbation is allowed to propagate through even mildly coupled dynamics.**

Four qualifications on this result, added per external review, that the above should not be read to overreach beyond: 1. **Repeated injection is not required for the qualitative phenomenon, though the precise mean-difference sequence differs.** A single, one-time initial-state perturbation produces the sequence of mean differences $CA^k \Delta_0$ at each step k , whereas persistent forcing produces the *cumulative sum* $C \sum_{j=0}^k A^j \Delta_0$ — these are different sequences, not the same one, a distinction an earlier version of this sentence blurred by calling them “the same sequence.” **What the two cases do share is the identical all-time unobservability condition:** if $CA^j \Delta_0 = 0$ for every $j = 0, \dots, n-1$ (which, by Cayley-Hamilton, extends to every $j \geq 0$), then every term in *both* sequences vanishes, so the qualitative phenomenon (confounding surviving feedback) requires the same condition either way — the persistent, repeated injection used in Section 3.2 is a modeling choice affecting the specific trajectory of mean differences, not a necessary ingredient of the unobservability condition

itself. 2. **“Generic” does not mean “typical of deployed markets.”** A randomly-drawn A is observable with probability 1, but real, structured systems can have unobservable components *by construction* — the claim that realistic market dynamics behave like a generic matrix is an assumption, not a derived fact, and we have not established it. 3. **Exact observability does not imply useful discrimination over a finite, noisy horizon.** For example, $C = (1, 0)$ and $A = \begin{pmatrix} 0.6 & \epsilon \\ 0 & 0.6 \end{pmatrix}$ is observable for every $\epsilon \neq 0$, yet a hidden difference in the second coordinate becomes arbitrarily hard to detect in finite time as $\epsilon \rightarrow 0$. Theorem 2 characterizes an asymptotic, all-time property; it does not by itself bound how many time steps are needed to exploit observability in practice. 4. **Identical state effects can still represent different causes.** $\Delta_0 = 0$ (the trivial, fully robust case) is mathematically degenerate but operationally significant: a fault and a compromised process can produce literally the same downstream state trajectory while differing in what they represent (authorized versus unauthorized). Observability of the state’s evolution does not by itself identify intent, authorization, or legal status — a distinction the identifiability framework in this paper is not built to make and does not claim to.

Why this matters for Section 7. This result requires no nonlinearity, no multi-agent interaction, and no ABIDES at all to demonstrate — it is a fact about the *linear* model itself, once a more realistic injection semantics (state-level rather than output-level) is used. It gives a cleaner, more fundamental candidate explanation for part of Section 7’s instability than feature choice or incident footprint alone: a real incident’s behavioral change plausibly propagates through system state (order flow affects the book, which affects future order flow), which is structurally a state-level disturbance, and Corollary 3 says such disturbances are confounding-fragile by default, robustly so only in special, non-generic dynamics. Section 7.8 later attempts to test this prediction directly in the ABIDES setting; the initial persistence ablation showed a large effect in the predicted direction, but external review correctly identified that this design conflates persistence with incident duration and cumulative activity, so it does not cleanly isolate the mechanism, and a follow-up isolation redesign (Section 7.10.2) remains inconclusive at the sample size tested. We therefore state this section’s contribution at the strength it actually has: a proven, exact mechanism *within the linear model*, directionally corroborated but not yet cleanly isolated in the nonlinear setting.

5. Pilot-scale empirical validation

We instantiate the model at small scale ($n = 3$ latent dimensions, $m = 2$ observed features, $T = 40$ steps per episode, $AR(1)$ dynamics) to confirm the theorem’s predictions computationally before relying on them. This is a **deliberately minimal, synthetic instantiation** — a proof of concept that the theory is internally consistent and empirically detectable in simulation — not a market-microstructure-realistic study; Section 9 specifies what is needed to scale it up.

Two regimes are constructed by choice of injection directions b_K, b_F : - **CONFOUNDED**: b_K, b_F chosen (via the observation matrix’s null space) so that $Cb_K = Cb_F$ exactly, satisfying (*). - **SEPARATED**: b_K, b_F chosen so that Cb_K, Cb_F are linearly independent.

A logistic-regression classifier is trained on simple trajectory summary statistics (mean, standard deviation, and net drift of each observed feature) to distinguish compromise from fault episodes, in both regimes, at two injection magnitudes.

Table 1. Passive classifier discrimination, by regime and magnitude (n=560 held-out episodes per cell).

Regime	Magnitude	Accuracy	AUC
CONFOUNDED (theorem: indistinguishable)	0.6	0.529	0.519
SEPARATED (theorem: distinguishable)	0.6	0.938	0.983
CONFOUNDED (theorem: indistinguishable)	1.2	0.488	0.485
SEPARATED (theorem: distinguishable)	1.2	1.000	1.000

As predicted by Theorem 1 and Proposition 1: in the CONFOUNDED regime, classifier accuracy sits at chance level (≈ 0.50) **regardless of injection magnitude** — more severe incidents are not easier to classify when the confounding condition holds, exactly because the two hypotheses remain observationally identical at any magnitude satisfying (*). In the SEPARATED regime, accuracy is high and *improves further* with magnitude, as Corollary 2 predicts via the effective signal-to-noise ratio.

5.1. Robustness to classifier choice

Proposition 1 guarantees the chance-level cap holds for *any* passive classifier under confounding, not just logistic regression. We confirm this against three additional, structurally different classifier families — a random forest (a flexible nonparametric model capable of fitting complex nonlinear decision boundaries within the limits of its finite depth and training data, not a literal universal approximator), Gaussian naive Bayes, and a naive single-feature threshold heuristic (mimicking a simple real-world detector) — using the same features and train/test split.

Table 1b. Classifier robustness check, CONFOUNDED vs. SEPARATED (n=420 held-out episodes).

Classifier	CONFOUNDED	CONFOUNDED	SEPARATED	SEPARATED
	accuracy	AUC	accuracy	AUC
Logistic Regression	0.474	0.480	1.000	1.000
Random Forest	0.488	0.497	1.000	1.000
Gaussian Naive Bayes	0.486	0.467	1.000	1.000
Threshold Rule (heuristic)	0.481	0.475	0.814	0.915

Every classifier family — including the random forest, which is flexible enough to fit complex nonlinear patterns in the training data if any existed — is pinned at chance level (≈ 0.47 – 0.49 accuracy, ≈ 0.47 – 0.50 AUC) under confounding, and every classifier substantially exceeds chance under separation. This confirms the chance-level result is a property of the confounding condition itself (as Proposition 1 proves, and which holds for *any* classifier, not only the flexible ones tested here), not an artifact of logistic regression’s specific inductive bias. The threshold heuristic is the weakest performer under separation (0.814 vs. ≈ 1.0 for the others), as expected of a single-feature rule, but still comfortably above chance — showing separation is detectable even by a naive real-world-style detector once the confounding condition does not hold. (Accuracy in this table uses each classifier’s standard 0.5

decision boundary, the conventional choice for a classification-quality metric — **corrected here per external review: AUC does not use any single threshold, since it is computed by ranking continuous scores across all possible thresholds; an earlier version of this parenthetical incorrectly said both metrics use the 0.5 boundary.** Section 6 below uses the cost-optimal threshold specific to the loss structure, generally different from 0.5, for the decision-cost analysis.)

6. Cost-of-ambiguity quantification

We attach a cost structure — containment cost $C_{OP} = 3$, unresolved-harm cost $HARM = 10$, needless-disruption cost $DISRUPT = 6$ (toy units, satisfying the corrected Proposition 1’s required $HARM > C_{OP}$ and $DISRUPT > 0$; not calibrated to any real deployment) — and compare policies across a sweep of the prior $P(K)$.

Terminology, corrected per external review. Three distinct risk quantities must be kept separate, in the ordering $R_{\text{oracle}} \leq R_Y^* \leq R_{\text{const}}$, where $R_{\text{oracle}} = \mathbb{E}[\min_a L(a, M)]$ is the true-cause oracle, $R_Y^* = \inf_{\pi} \mathbb{E}[L(\pi(Y), M)]$ is the best achievable policy using only the observation Y , and $R_{\text{const}} = \min_a \mathbb{E}[L(a, M)]$ is the best *constant*-action policy. **An earlier version of this section conflated R_Y^* with R_{const} in both regimes.** Exact confounding is a *sufficient*, not necessary, condition for this equality: under $(*)$, Y carries no information, so $R_Y^* = R_{\text{const}}$ trivially. But the two can also coincide even when Y is informative, if the available information is never strong enough to flip the loss-optimal action across the relevant range of the observation — for example, at a low prior p , a moderately informative Y can still leave both posteriors $P(K \mid Y=0)$ and $P(K \mid Y=1)$ below the containment threshold, so the optimal action never changes and $R_Y^* = R_{\text{const}}$ despite Y carrying real information. We do not claim to have distinguished these two cases empirically in this paper’s simulations; the point is definitional, to avoid treating an absence of measured cost benefit as proof that the underlying observation laws are equal. Under separation, in our specific simulated instances, R_Y^* can be — and empirically is — substantially *below* R_{const} ; treating R_{const} as “the optimum” in that regime was a labeling error, not merely an imprecise phrase, and every table and claim below is corrected accordingly.

A second, more consequential correction: a threshold-calibration bug invalidated a headline finding from an earlier draft. The classifier is trained on a *balanced* (50/50) dataset, so its raw output $s(y)$ estimates $P(K \mid Y = y)$ under a training prior $p_{\text{train}} = 0.5$, not the deployment prior p . An earlier version of this section applied the deployment-calibrated threshold $q^* = \text{DISRUPT}/(\text{HARM} - C_{OP} + \text{DISRUPT})$ directly to this raw score — a prior-shift error of exactly the kind Elkan (2001) documents for cost-sensitive learning. We verified computationally that this is not a minor effect: at $P(K) = 0.05$, the deployment threshold is $q^* \approx 0.462$, while the correctly prior-adjusted threshold on a balanced-training score is $s^* \approx 0.942$ — since an uninformative classifier’s raw score hovers near 0.5, comparing it to 0.462 makes it mostly predict “contain,” while the correct comparison point 0.942 makes it mostly and correctly predict “leave.” **Once this is fixed using the standard Bayes’ rule prior-correction** $q_p(y) = \frac{(p/p_{\text{train}})s(y)}{(p/p_{\text{train}})s(y) + \frac{1-p}{1-p_{\text{train}}}(1-s(y))}$, **the earlier finding that “a passive classifier under confounding can be actively worse than ignoring it” disappears entirely: our fitted classifier’s empirical cost becomes statistically indistinguishable from R_{const} (to within simulation noise) once correctly threshold-adjusted.** We are precise about what this does and does not establish, per external review: Proposition 1 characterizes the Bayes-optimal policy using Y (R_Y^*) as equal to R_{const} under exact confounding — it does not guarantee that any *given fitted* policy will exactly attain that value, since a learned score can retain residual estimation or calibration error even after correct prior adjustment, and such error can in principle still produce a policy strictly worse than R_{const} . What we have shown is empirical: our specific corrected classifier approximately attained the theoretical floor in this experiment, not that every classifier must. We withdraw the earlier finding rather than retain it alongside the correction, and we are explicit that this was a genuine error, not a restatement: the previous headline was wrong, not merely imprecisely worded.

We compare four policies: **Fixed** (always contain), **Leave** (always leave alone), **Classifier** (the passive classifier’s prior-corrected posterior, thresholded at q^*), and **Oracle**.

Table 2 (corrected). Expected cost per episode (toy units), CONFOUNDED regime.

$P(K)$	Fixed	Leave	Classifier (prior-corrected)	Oracle	R_{const} (best constant)	R_Y^*	Classifier excess over R_Y^*
0.05	5.85	0.50	0.50	0.15	0.50	0.50	0.00
0.10	5.70	1.00	1.00	0.30	1.00	1.00	0.00
0.20	5.40	2.00	2.00	0.60	2.00	2.00	0.00
0.30	5.10	3.00	3.00	0.90	3.00	3.00	0.00
0.50	4.50	5.00	4.45	1.50	4.50	4.50	−0.05

Under confounding, $R_Y^* = R_{\text{const}}$ exactly (Proposition 1), so the “classifier excess” column above is a direct empirical check of whether the prior-corrected classifier matches this floor — and it does, to within simulation noise at every prior (the -0.05 at $P(K) = 0.5$ is sampling variance, not a systematic effect, and is not distinguishable from zero given the sample size used).

Table 3 (corrected). Expected cost per episode (toy units), SEPARATED regime.

$P(K)$	Fixed	Leave	Classifier (prior-corrected)	Oracle	R_{const} (best constant, not the achievable optimum here)	Classifier excess over R_{const}
0.05	5.85	0.50	0.26	0.15	0.50	−0.24
0.10	5.70	1.00	0.43	0.30	1.00	−0.57
0.20	5.40	2.00	0.68	0.60	2.00	−1.32
0.30	5.10	3.00	1.01	0.90	3.00	−1.99
0.50	4.50	5.00	1.52	1.50	4.50	−2.98

Here R_{const} is *not* the achievable optimum — Y is informative, so R_Y^* is genuinely lower, and we have not computed its closed form for this simulated

(non-Gaussian-summary) setting, only the analytical Gaussian benchmark in Section 4’s Corollary 2 discussion for the raw trajectory. The “classifier excess over R_{const} ” column shows the classifier substantially beats the constant-action baseline and sits close to the oracle, which is a legitimate and correctly-labeled finding — but it should not be read as “excess over the optimum,” since R_{const} is not the optimum in this regime.

Every “Fixed,” “Leave,” and “Oracle” value above matches the exact closed-form values $pC_{OP} + (1 - p)\text{DISRUPT}$, $p\text{HARM}$, and pC_{OP} respectively, to full displayed precision — a direct, checkable correctness test confirming the resampling fix from the prior round remains valid; the prior-shift bug identified here is a separate, additional error in how the classifier’s threshold was applied, not a recurrence of the earlier resampling issue.

The corrected, honest findings, replacing both the original and the (also incorrect) intermediate versions:

1. **Neither constant policy is good away from its favorable regime.** Comparing Fixed against Leave shows an operator who always contains does badly when attacks are rare, and an operator who never contains does badly when they are common — this remains a valid critique of any fixed, non-adaptive policy, unaffected by the corrections above.
2. **Under separation, a properly prior-corrected classifier substantially beats the constant-action baseline and sits close to the oracle** — a legitimate, correctly-labeled finding once “excess over R_{const} ” is not conflated with “excess over the optimum.”
3. **Under confounding, our properly prior-corrected fitted classifier empirically matched the constant-action baseline, consistent with (but not logically guaranteed by) Proposition 1.** Proposition 1 establishes that the *Bayes-optimal* policy using Y equals the constant-action baseline under exact confounding — it does not guarantee that any given fitted policy will exactly attain this, since residual estimation or calibration error can in principle still produce a classifier strictly worse than the constant baseline even after correct prior adjustment. Our corrected classifier happened to approximately attain the floor in this experiment; we report this as an empirical finding, not as a restated theorem. The earlier claim that it could be *actively worse* is retracted: it was an artifact of applying a deployment-calibrated threshold to a training-prior-calibrated score, not a property of confounding itself. **The genuinely useful lesson from this whole**

episode is methodological, not substantive: cost-sensitive decision rules built on classifiers trained at one class balance and deployed at another must correct for that prior shift explicitly, or they can appear to perform arbitrarily badly for reasons that have nothing to do with the underlying identifiability limit — a caution we did not originally appreciate ourselves and are glad to have had identified.

6.1. Sensitivity to the cost triple (corrected, with prior-correction applied)

Because $(C_{OP}, \text{DISRUPT}, \text{HARM}) = (3, 6, 10)$ was chosen only to be strictly ordered and asymmetric, we re-ran the cost calculation of this section across seven triples spanning different ratios and overall scales — six genuinely additional triples plus the original $(3, 6, 10)$ triple repeated here for direct comparison — at a fixed representative prior $P(K) = 0.15$, **using the fully corrected pipeline (bug-fixed resampling, and — following this round’s correction — the prior-corrected threshold, not the deployment threshold applied to a training-prior-calibrated score).**

Table 2b (corrected). Cost-of-ambiguity robustness across cost triples.

C_{OP}	DISRUPT	HARM	Classifier excess over R_{const} , CON- FOUNDED	Classifier excess over R_{const} , SEP- ARATED
1	2	5	0.000	−0.566
3	6	10	0.000	−0.966
1	4	10	0.000	−1.286
2	3	15	0.029	−1.950
1	6	10	0.000	−1.242
5	8	12	0.000	−0.823
1	1.5	3	0.000	−0.282

Across every triple tested, once the prior-correction is applied, **CONFOUNDED-regime excess over R_{const} is exactly zero (to within simulation noise) in 6 of 7 triples**, and negligibly small (0.029) in the seventh — confirming, robustly, that Proposition 1’s guarantee holds for the constant-action floor and that the earlier “actively worse than a constant policy” finding

does not survive correction at any tested cost triple, not merely the original one. **SEPARATED-regime excess is consistently and substantially negative across every triple** (classifier beats the constant-action baseline everywhere signal exists), confirming that finding’s robustness as well. This table supersedes two earlier, now-withdrawn versions: an original uncorrected version (CONFOUNDED values 0.32–4.72) reflecting the resampling bug, and an intermediate version (CONFOUNDED values 0.322–3.723) that had fixed the resampling bug but still applied the deployment threshold without the prior-shift correction — an internal duplication error where both versions were briefly left in this document simultaneously, now resolved by keeping only the fully corrected table above. As in Section 6, $\text{CoA}^* = \min\{(1 - p)\text{DISRUPT}, p(\text{HARM} - C_{OP})\}$ remains an exact, deterministic function of the cost triple and prior alone (no dependence on any classifier or data), included in Section 6’s discussion as context for checking the “excess” columns against that floor, not as a quantity requiring its own empirical replication here.

6.2. *An illustrative dollar-scale reconstruction (not an empirical calibration)*

The cost constants used above are illustrative toy values. To give a sense of scale in real terms, we reconstruct a comparable triple from the Almgren–Chriss market-impact framework (linear temporary-plus-permanent impact, $L(X, T) = \epsilon X + \frac{\gamma}{2} X^2 + \eta X^2/T$), using the framework’s own original illustrative parameters (a \$50 stock, 5M-share average daily volume, \$0.125 spread, and the paper’s example impact coefficients) applied to three representative scenario sizes: a normal containment action (50,000 shares over 5 minutes), a needlessly disrupted benign order (the same size, forced into an urgent 5-minute unwind rather than a patient 3-hour one), and an unmitigated adversarial sequence (200,000 shares over a 30-minute exposure window). **We are explicit that this is an arithmetic reconstruction of an illustrative scenario using Almgren–Chriss’s own example parameters, not an empirical calibration to any measured trading incident, market, or venue** — the earlier draft’s use of “calibrated” overstated what this is, and we correct that here. In particular, the assumed containment trade (50,000 shares in 5 minutes) is roughly 78% of expected volume in that window under a uniform intraday-volume assumption — an aggressive execution schedule whose realism directly affects every downstream number, and which we have not validated against any specific real venue or instrument.

This reconstruction gives $(C_{OP}, \text{DISRUPT}, \text{HARM}) \approx (\$490.9\text{k}, \$474.0\text{k}, \$1.3175\text{M})$ — satisfying the corrected Proposition 1’s requirement $\text{HARM} > C_{OP}$, but with one feature worth flagging explicitly: **C_{OP} and DISRUPT are nearly equal, with C_{OP} very slightly larger**, so the toy triple’s intuitive story (“needless disruption costs more than routine containment”) is not what this reconstruction shows; the disruption figure here mainly reflects a forced-urgency liquidation premium on a benign order, comparable to, not larger than, the containment action’s own cost.

Applying the corrected Proposition 1 to this triple, at $P(K) = 0.05$: the oracle’s expected cost is $p C_{OP} \approx \$24,550$, the best-constant baseline is $\approx \$65,875$, and the *optimal ambiguity cost* — the theorem-guaranteed floor, $\min\{(1-p)\text{DISRUPT}, p(\text{HARM} - C_{OP})\}$ — is approximately **\$41,000** (precisely \$41,328, though we round to reflect the illustrative, not measured, nature of the inputs — an earlier draft’s \$41,450 reflected excess rounding precision in the intermediate cost triple and is corrected here per external review’s independent recalculation, and a still-earlier headline of \$188,000 was withdrawn several rounds ago as the artifact of a since-fixed resampling bug). We present this as an order-of-magnitude illustration only, not a measured or venue-representative figure, and any classifier’s *additional* regret beyond this floor must be measured with a properly prior-corrected policy (Section 6), not assumed equal to a naively-thresholded classifier’s total regret.

6.3. A corrected loss table for the audited fault mechanisms — a genuine finding, now with its own bug-fix story

Section 3.4 established that most of this paper’s simulated F mechanisms do not satisfy the “leave alone costs \$0” assumption their loss table requires — they submit real, market-affecting orders. We priced this directly using the project’s Almgren-Chriss impact formula. **An earlier version of this section reported “leave alone” costs of \$286K–\$21.8M and a headline claim that always-contain becomes cost-optimal in 7 of 8 audited cells — this has since been found to rest on a real methodological bug in our own costing, not a substantive finding, and is corrected here rather than left standing.**

The bug: the initial costing compressed F ’s full-day-equivalent trading volume into the 30-minute exposure window as if it were being executed as a single block trade, rather than costing only the volume F actually submits within that window. This overstated F ’s market impact by roughly two orders of magnitude. The Almgren-Chriss impact coefficients $(\eta, \gamma, \varepsilon)$

themselves were not the problem — they are legitimate, ADV/spread-derived constants for the stock in question and did not need rescaling; the fix was correcting the *volume figure* fed into them.

Corrected result: “leave alone” costs drop by roughly 130–165x across every design once this is fixed (e.g., V1’s confounded regime: \$21.8M → \$132.6K). **This specific corrected figure was then itself hand-verified against directly-measured simulated volumes, per external review, and found to still be wrong** — not from the window-compression bug above, but from a *third*, independent error: the formula mixed a 86,400-second calendar day with a 390-minute trading day, introducing a spurious 3.69x factor, exposed cleanly by a zero-noise design (V2) where formula and ground truth diverge with no measurement noise to blame (predicted 15,508 shares vs. an actual, directly-logged 4,235). A fourth check (V1) showed the formula’s 14% agreement with ground truth was coincidental — two separate, opposite-signed errors nearly canceling for that specific design’s parameters, not evidence the formula was actually correct. Two further spot checks (V3, V4) showed different, non-cancelling error ratios (1.6x over, 0.8x under), confirming the formula is unreliable in general, not merely imprecise for one design.

We therefore do not report the formula-derived dollar figures as final, and, per external review, we correct their description here too: these are model-based costs using measured submitted volume, not a direct measurement of economic loss. Recomputing “leave alone” costs directly from measured simulated order volumes (the incident agent’s own logged submitted-order actions, not executed volume — cancellations and partial fills are not accounted for) rather than the Almgren-Chriss formula’s *predicted* volume input: **the qualitative conclusion survives, and the measurement itself has been completed to full precision.** A proper sample (100 episodes per design/regime, 800 episodes total) of directly-measured *F*-mechanism *submitted* order volumes, costed using the same Almgren-Chriss impact coefficients (confirmed correct in themselves) but with measured rather than formula-predicted volume, gives the final table below. A cross-check against the smaller single-episode-batch hand-verification from the previous round confirms the larger sample’s means diverge by no more than 2.9% in any design — the earlier spot checks were representative, not fortunate.

Table 2c. “Leave alone” cost for *F*, from directly-measured order volume (100 episodes/cell).

Design	Regime	Mean shares	“Leave alone” cost	q^*
V1	Separated	30,902	\$33.09K	0.348
V1	Confounded	55,691	\$104.67K	0.309
V2	Separated	4,235 (deterministic)	\$0.85K	0.364
V2	Confounded	4,235 (deterministic)	\$0.85K	0.364
V3	Separated	6,547	\$1.81K	0.364
V3	Confounded	11,348	\$4.91K	0.362
V4	Separated	12,470	\$5.85K	0.362
V4	Confounded	34,611	\$41.24K	0.344

Update, per external review: this claim does not hold universally and is corrected here rather than left standing. An earlier version of this paragraph claimed all 40 (design $\times$ regime $\times$ prior) cells favor the classifier over always-contain. The full 50-cell audit in Section 7.7 (which includes Study 4, absent from this section’s original 40-cell scope) found **2 cells** — **Study 5, both regimes, at $P(K) = 0.50$** — **do not beat always-contain**, contradicting this section’s original claim even restricted to its own 40-cell scope. q^* itself remains a property of the cost structure alone, correctly independent of the deployment prior (it stays tightly clustered in $[0.309, 0.364]$ across every design and regime) — this internal-consistency property is unaffected by the correction above, which concerns how often the *classifier’s realized cost* falls below the always-contain baseline, not the threshold formula itself. See Section 7.7 for the complete, correct count (50 of 50 cells beat always-contain; 41 of 50 beat best-constant on the unified large-N basis, using the direct class-conditional weighting method that superseded the resampling-based one).

A precision issue in what this table actually computes, per external review, stated rather than left implicit — and a formula error in an earlier attempt to state this precisely, corrected here. The Almgren-Chriss impact function used throughout this paper includes both temporary and permanent impact at fixed duration T : $L(X, T) = \varepsilon X + \left(\frac{\gamma}{2} + \frac{\eta}{T}\right) X^2$. By Jensen’s inequality, since this is convex in X : $\mathbb{E}[L(X, T)] = L(\mathbb{E}[X], T) + \left(\frac{\gamma}{2} + \frac{\eta}{T}\right) \text{Var}(X)$. **An earlier version of this paragraph incorrectly omitted the temporary-impact term η/T from this formula, writing only $\frac{\gamma}{2}\text{Var}(X)$** — a real error, not a simplification, since it directly contradicted the worked numerical example given in the same para-

graph (which used the full, correct coefficient). Pricing the *mean* measured volume, as this table does, is not the same as the *mean* of the per-episode costs, and understates the latter whenever volume varies episode-to-episode (the deterministic V2 design is exempt from this specific gap, since its volume has zero variance by construction).

Var(X) has since been measured directly from the existing 100-episode samples for all three stochastic designs, closing the gap the previous version left as an illustrative hypothetical. Using the confirmed-correct combined coefficient $\frac{\gamma}{2} + \frac{\eta}{T} = 0.000032625$:

Design/register	Mean shares	SD shares	Naive (mean- volume) cost	Jensen's- gap correc- tion	Corrected cost	Gap as % of naive
V1 sepa- rated	30,902	1,670	\$31,154.87	+\$90.95	\$31,245.82	0.29%
V1 con- founded	55,691	2,334	\$101,184.98	+\$177.72	\$101,362.70	0.18%
V3 sepa- rated	6,547	475	\$1,398.48	+\$7.36	\$1,405.83	0.53%
V3 con- founded	11,348	627	\$4,201.26	+\$12.83	\$4,214.10	0.31%
V4 sepa- rated	12,470	899	\$5,073.44	+\$26.34	\$5,099.79	0.52%
V4 con- founded	34,611	2,895	\$39,081.05	+\$273.51	\$39,354.57	0.70%

The gap is small in every measured case (0.18%–0.70% of the naive quadratic cost) but consistently nonzero and consistently in the same direction (understating cost), confirming the earlier illustrative estimate’s order of magnitude was correct even though it was not yet backed by actual measurement. This table should be read as “the cost of executing the mean measured volume, at fixed duration T , corrected for episode-to-episode variance,” which is a well-defined and useful quantity in its own right, not as a claim to have captured every source of episode-level cost variation; for variable duration the expectation involves X^2/T and does not reduce to this fixed- T identity without further

assumptions.

We report this pattern plainly because it is now a recurring one in this paper’s own history, not because it is unique to this finding — and this is the third instance of a costing formula producing a plausible-looking but wrong number in this section alone. A striking, headline-worthy result (first, “a passive classifier can be actively worse than a constant policy” in Section 6; then, “leaving a fault alone is actually enormously expensive, flipping the optimal policy”; then, a formula-derived correction to that second finding that was itself wrong twice over) has repeatedly turned out to rest on a formula or calibration bug rather than a property of the underlying system. The genuinely durable finding across all three episodes is not any specific number, but the meta-lesson: **a plausible-looking formula-derived number should be checked against direct measurement before being trusted, not merely checked for internal consistency** — a standard this section now meets in full, for both its qualitative conclusion and its precise figures.

What remains true from Section 3.4’s audit, unaffected by this correction: F ’s submitted order volume is not literally free to price at \$0 under this model — the final, measured-volume-based costs (\$0.85K–\$104.67K, per the table above) are still nonzero and still larger than the original loss table’s assumed \$0 — so the loss table in Sections 6–7 remains an illustrative simplification, not a validated economic costing, of what these specific simulated F mechanisms actually do or the incremental harm they cause relative to a baseline. (We correct a stale reference here too: an earlier version of this closing paragraph cited the superseded, still-buggy “\$132.6K-scale” figure from the second correction attempt rather than the final table’s \$104.67K for that same cell — a leftover from before the third bug was found, now fixed.) **What is now corrected is the *consequence*, at the accurate strength:** per the full 50-cell audit (Section 7.7, using the direct class-conditional weighting method), this simplification leaves the classifier ahead of always-contain in all 50 cells and ahead of the best constant policy in 41 of 50 — a strong but not universal pattern, with the 9 exceptions all low-prior cells (6 of them Study 5) where always-leave dominates — so Sections 6–7’s classifier-value claims stand for four of the five profiles, with Study 5’s own status now precisely characterized rather than folded into an “unchanged in every configuration” overclaim an earlier version of this section made.

7. An empirical study of observation matching in a realistic non-linear market

A critical reproducibility disclosure, added after this section was substantially complete and placed here so it is read before anything else, and updated once more as the investigation deepened. A follow-up RNG audit (Section 7.10.5) found and patched **two further unseeded-RNG bugs** beyond the `get_wake_time()` issue originally disclosed, and confirmed bit-identical cross-process reproducibility only *after* all three were fixed. **A subsequent, still deeper investigation then found a fourth bug: worker processes cache their first lazy import of the patched code for their entire lifetime, so results generated after the on-disk patch could still silently reflect pre-patch behavior depending on a worker’s exact import timing — a failure mode invisible to file-timestamp-based provenance checks.** This directly affected the frozen confirmation set discussed below, which we previously and incorrectly described as fully clean; **that description is withdrawn.** A genuinely verified-clean pipeline (forced re-import per task, plus a runtime behavioral fingerprint confirming all three patches are active, independent of timestamps) has since been built. **Essentially every ABIDES result reported in this section — Sections 7.1 through 7.8 — was generated before verified-clean generation was available, and should be understood as statistically valid (genuine, if not fully deterministic, randomness) but not confirmed bit-for-bit or provenance-clean by the standard we now hold results to.** Any claim in Sections 7.1–7.8 of the form “reproduced with seed X ” or “generated after the fix” should be read with this caveat, and a submission-ready version of this paper should regenerate the full result set under the verified-clean pipeline rather than rely on these runs. We disclose this prominently, and are revising it again as the investigation progresses, because it is exactly the kind of caveat that is easy to bury and important not to — including the uncomfortable fact that our own first attempt at this disclosure was itself incomplete.

A framing correction, made explicit before anything else in this section, per external review. Everything that follows was previously described using language like “the theorem’s prediction inverted” or “the quantitative floor was violated.” That framing is incorrect and is withdrawn. Theorem 1 concerns equality of the *entire* observation trajectory law, $P_K(Y_{0:T}) = P_F(Y_{0:T})$. What the constructions below actually do is

match the two mechanisms on *one or a few chosen summary statistics* (mean price drift, order count, mean spread) while other observable channels (order counts, depth, cancellation rate, size) remain measurably different. **Matching one moment is not equality of the joint trajectory distribution, and a classifier exploiting an unmatched channel does not contradict Theorem 1 — this would be true even in a fully linear system.** A trivial counterexample makes this concrete: $Y \mid K \sim \mathcal{N}((0, 2), I)$, $Y \mid F \sim \mathcal{N}((0, -2), I)$ have identical first-coordinate means but a highly informative second coordinate; no nonlinearity is needed for this pattern.

Accordingly, we relabel the constructions below **summary-matched** (previously “CONFOUNDED”: matched on one or more chosen statistics) and **contrast** (previously “SEPARATED”: deliberately given a large chosen-attribute difference), and we do not claim any comparison between them is a test of Theorem 1’s prediction. What this section actually establishes — and this is still a real, useful finding — is an empirical account of **which observable channels leak information under which summary-matching choices**, and how sensitive that leakage is to feature selection, calibration target, classifier choice, and incident footprint. Section 7.3’s synthesis and Section 7.7’s practitioner checklist are reinterpreted below in these corrected terms. We also note, following the same external review, a genuine ABIDES-based **negative control**: two mechanisms with *identical* observable order-generation kernels (not merely matched on one statistic) should show no repeatable discriminatory signal under valid, independently-generated evaluation data; a persistent signal there would indicate an implementation or evaluation-boundary problem. **This control has since been run and finalized (Section 7.10.1)** — an earlier version of this paragraph described it as not-yet-run future work, which is now stale; see Section 7.10.1 for the full result, including a genuine diagnostic journey (an initial pass, a concerning intermediate reading, a real seed-block issue found and fixed, and a final result with exact interval bounds now computed two ways: accuracy passes a genuine equivalence test against the pre-specified margin, AUC’s point estimate is inside the margin but its interval is not — we correct a stale claim here too, from an earlier version of this paragraph that said “we do not have the interval bounds needed,” which is no longer accurate).

The linear model’s Theorem 1 remains exact and, per Sections 5–6, reproducible within its own stated scope. This section does not test whether Theorem 1 is true — it is proved, not tested — nor does it establish a validated “linearization” bridge from Theorem 1’s model to ABIDES’s dynamics

(no operating point, Jacobian, or approximation-error bound connecting the two has been derived; that would be required to make a linearization claim, and we have not made one). What we did was inject compromise/fault mechanisms as real order attributes (submission rate, order size, price aggressiveness, directional bias, cancel/replace churn) into ABIDES, a real discrete-event multi-agent market simulator, and study the resulting classification and cost behavior as an object of independent empirical interest — a comparison of a stylized linear model and an agent-based one, not a validation or falsification of the former by the latter.

7.1. First result: summary-matching on one statistic leaves other channels informative

Matching mean price drift alone does not come close to matching other observables. At the calibrated magnitude, the two mechanisms’ mean order counts differed by roughly 2x, cascading into large gaps in depth and spread statistics the calibration search did not control. Two independent calibration searches, differing only in random seed, selected different “best” magnitudes for even this one matched statistic — the discovery process itself is noisy.

A passive classifier discriminated the summary-matched scenario more accurately than the contrast scenario. 82.9% accuracy (AUC 0.921) on the summary-matched construction, versus 69.5% accuracy (AUC 0.733) on the contrast construction — the opposite ranking from the pilot’s Table 1 pattern, and worth investigating for that reason, but **not, on its own, evidence against Theorem 1**, since the summary-matched construction was never shown to satisfy Theorem 1’s actual premise. **Both figures were later independently re-verified at large N (1000 episodes/class, 2000 total), after the contrast-regime accuracy showed concerning instability across smaller-sample regeneration attempts made in response to a provenance-verification issue found elsewhere in this section (0.695 $\rightarrow$ 0.633 $\rightarrow$ 0.417 across three regenerations); the large-N run confirmed the original 69.5% figure was correct all along and the intermediate low readings were small-sample noise from an atypical draw, not evidence of an unstable or fake effect. The summary-matched regime’s large-N recheck came back stronger, not weaker: 89% accuracy versus the original single-run 82.9% — meaning the reversal gap reported in this section understates, if anything, the fully-powered result. The cost-of-ambiguity**

quantification (once corrected — see Section 6) inherits the same ranking; this too should be read as “the summary-matched scenario as constructed left more decision-relevant information available to this specific policy than the contrast scenario did” — **correcting an earlier version of this sentence, which incorrectly attributed the additional information to the contrast scenario, backwards from what the accuracy figures above actually show (summary-matched is the more accurately classified regime here, not contrast)** — not as a violation of any theorem.

This first result raised an obvious question: is this ranking a robust property of these two specific constructions, or an artifact of *this particular* implementation’s choices? Section 7.2 answers this with a systematic sensitivity analysis, deliberately designed so that no single run could be mistaken for the final word.

7.2. Sensitivity analysis: five extension studies

Study 1 — Statistical robustness of the accuracy ranking. Five independent seeds confirm the accuracy/AUC ranking (summary-matched more classifiable than contrast) is not noise: bootstrap 95% CI on the accuracy gap is $[+0.154, +0.192]$, on the AUC gap $[+0.199, +0.269]$. The cost-metric version of this ranking is directionally consistent but noisier (2 of 25 seed $\times$ prior combinations flip sign at extreme priors) — and, following the Section 6 correction, should now be re-examined using the corrected, non-buggy cost pipeline and the classifier-excess-over-best-constant decomposition rather than the earlier conflated figure; this re-analysis has not yet been done for the ABIDES-side cost tables (Section 9 lists it as outstanding).

Study 2 — Classifier-family sensitivity. The ranking holds for three of four classifier families: logistic regression, Gaussian naive Bayes, and random forest (0.895 vs. 0.752). The fourth, a naive single-feature threshold rule, initially produced 0.343 accuracy — below chance, the signature of an inverted decision boundary, traced to a hardcoded-direction bug (fixed: 0.657 vs. 0.705, a mild dissent from the other three classifiers’ ranking, via a properly-oriented rule). We read this as evidence that the ranking’s strength depends on classifier capacity to exploit whatever unmatched channels are present, not as evidence about any theorem.

Study 3 — Population and scale sensitivity. The ranking holds across a roughly 4x range of background population size and at double session length, ruling out fragility to one arbitrary population scale.

Study 4 — Feature-set sensitivity (the pivotal result for this section). Adding order-flow imbalance, signed executed volume, and queue concentration **reverses the ranking again:** the contrast scenario’s accuracy reaches 0.996 (CI [0.988, 1.00]), the summary-matched scenario sits at 0.817 (CI [0.750, 0.883]), both stable under replication. **Correct interpretation:** the contrast scenario was, in fact, always highly distinguishable on the attributes it was designed to differ on — the original 4-channel feature set simply failed to expose that difference, while an unrelated activity-level mismatch in the summary-matched scenario happened to leak through those same 4 channels more strongly. Adding the right features corrects the *feature-access* problem, not a property of either scenario’s underlying observation law. One data-availability note, corrected here per external review and an earlier VM finding that was never propagated to this specific sentence: **an earlier version of this paragraph claimed “ABIDES does not retain per-trade execution price for ordinary fills,” ruling out a literal volume-weighted-average-price feature — this is factually wrong and is corrected here.** A dedicated check confirmed execution prices are logged for 100% of tested trade bins. **We also withdraw a related, unsupported claim from an earlier version of this paragraph,** which asserted that a VWAP-feature comparison exists in Section 7.10.6 and found negligible difference from the signed-volume proxy — Section 7.10.6 contains only the signed-volume sign-convention discussion, no VWAP comparison, sample size, or numerical outcome, and “VWAP” appeared nowhere else in that section. Either that comparison should be run and reported with its own sample size and result, or the claim should not be made; we do not repeat it here. The signed-volume proxy used in this study remains a reasonable feature choice on its own merits, independent of any VWAP comparison.

Study 5 — A second, independently designed injection construction. Built from the attribute specification alone, using a different attribute subset, a deterministic wake-up clock, and a different calibration target. Five-seed replication: the summary-matched case sits at chance (accuracy CI [0.446, 0.608]), the contrast case shows a small but statistically distinguishable signal (AUC CI [0.532, 0.633], excluding 0.5). **A caution from external review applies directly here:** a confidence interval crossing 0.5 means this specific tested procedure did not detect a reliable departure from chance — it does not establish that the two mechanisms’ observation laws are equal, and the interval [0.446, 0.608] leaves room for a practically meaningful amount of undetected signal. We use “no reliable discrimination

detected by the tested classifiers,” not “confirmed indistinguishable,” from here on. **This caution was vindicated by later, larger-scale verification:** the contrast case’s signal, initially read as “small,” was substantially underestimated. Two small-sample estimates (0.522, then 0.567 accuracy) both undershot a subsequent large-N run (1000 episodes/class): accuracy 0.615, confirmed tightly stable by a 30-split check (SD 0.018, comparable to the stability standard used elsewhere in this paper) rather than another unstable small-sample artifact. **A final end-to-end regeneration under the fully-hardened pipeline (Section 7.10.9) confirms this upward correction and refines the AUC figure:** accuracy 0.615, AUC 0.661 (the earlier 0.652 AUC estimate came from a pre-hardening run; the hardened figure supersedes it). Table 5 now reports these hardened values throughout.

Terminology note for Sections 7.3–7.10 below: the labels **CONFOUNDED** and **SEPARATED** appear throughout the remaining subsections as inherited shorthand for, respectively, the summary-matched and contrast constructions defined above. Neither label should be read as a claim that Theorem 1’s premise (full trajectory-distribution equality) was established or tested for either construction — they denote specific engineering choices (which statistic was matched, which was left free to differ). **The paragraphs below have been rewritten, not merely annotated, to remove language asserting that any result “violates” the theorem, that a “quantitative guarantee” was tested and failed, or that the theorem’s floor “should have” transferred — external review correctly identified that a terminology note alone does not repair a sentence that still asserts the thing being disclaimed, so those sentences are corrected directly here rather than left standing.**

7.3. Synthesis: what these constructions show about observation-matching, not about Theorem 1

Three genuinely different classification profiles emerged from three reasonable implementation choices — the original construction (accuracy ranking one direction, confirmed with high statistical confidence), a richer-feature version of the same construction (ranking reversed, also confirmed with high statistical confidence), and an independently designed injection layer (near-chance discrimination in one scenario, a weak but statistically real signal in the other). A fourth data point, the corrected threshold-rule baseline in Study 2, adds that even *within* one fixed construction, a weaker classifier can land on a different ranking than stronger ones do on the same data. We do

not treat any single one of these as *the* answer; the pattern across all four — that the ranking flips depending on engineering choices with no analog in the linear model — is the finding.

The mechanism is structural, and it concerns observation-matching difficulty, not theorem validity. In the linear model, one vector equality (in $\mathbb{R}^m$, the observation dimension — not literally “one scalar equality” unless the operator has only a single observable channel, a precision worth stating exactly rather than loosely) zeroes out the difference between mechanisms in *every* observed coordinate at *every* time step simultaneously, because C is fixed, known, and identical across time, and — per Section 4.1’s Theorem 2 — that equality is itself feedback-fragile except under non-generic dynamics. In a nonlinear multi-agent market, there is no single fixed linear map to match in the first place: an engineer attempting to construct an analog of exact confounding must choose a finite set of statistics to match (mean drift, order count, spread), and whatever is left unmatched can leak. Section 7.2’s Studies 4 and 5 (with confidence intervals, not single-run estimates) show this leakage depends on at least two further choices — **which, per a direct audit, are representable but simply unexplored in this paper’s linear-model pilot, not structurally absent from the linear model as an earlier version of this section claimed: the richness of the observer’s feature set** has a direct linear-model analog in the number of rows of C (how many gateway-observable channels the operator has), which was never expanded beyond its original two rows in any linear-model script in this project, even though ABIDES’s analogous feature-richness axis (Study 4) was explicitly swept; and **the incident’s footprint size relative to the background population** has a functional analog in the ratio of injection magnitude to the process/observation noise scale (Q, Σ), which the linear-model pilot swept the numerator of (Table 1’s magnitude sweep) but never the denominator of — the noise scale was hardcoded and identical across every linear-model script in this project. **A third factor, from the corrected Study 2, is classifier capacity** — and here an earlier version of this paragraph made a claim that Section 7.2’s own results contradict: it asserted “no ABIDES result in this project was ever checked against more than logistic regression,” concluding classifier choice was more explored on the linear side. **This is false and is corrected here per external review:** Study 2 (Section 7.2) explicitly tested four classifier families on ABIDES data (logistic regression, Gaussian naive Bayes, random forest, and a threshold rule) — the same count as the linear-model pilot’s Table 1b. Classifier choice is, in fact,

similarly explored on both sides, not asymmetrically so in either direction; the earlier claim was a genuine reporting error, not a subtler distinction we failed to state precisely. **The one factor genuinely absent from the linear model, not merely unexplored, is the calibration-by-search mechanism ABIDES’s summary-matched construction needs** — the linear model’s exact confounded case is constructed analytically via C ’s nullspace, with no search required, because linearity admits a closed-form solution; ABIDES’s nonlinear mapping from attributes to summary statistics has no such closed form, so its construction genuinely requires an approximate grid search the linear model structurally does not need. We have not disentangled how much of this instability traces to Theorem 2’s state-feedback mechanism specifically versus these factors — Section 7.8’s ablation attempted this and remains inconclusive (Section 7.10.2) — and separating their relative contributions is a natural next step, not a claim we make here.

What the evidence actually supports. Across every tested construction, genuine chance-level discrimination (consistent with Theorem 1’s confounding prediction) appeared in one case: the one-off ablation (Section 7.8), where reduced persistence, duration, and cumulative activity together — not state-feedback alone, per that section’s own correction — brought the summary-matched case down to a level consistent with chance. **Study 5’s summary-matched case was previously also cited here as a second such case, attributed to “footprint too small to register” — that attribution is retracted, matching Section 7.5’s own later, dedicated sweep, which found footprint alone does not explain Study 5’s behavior; we do not have a clean isolated explanation for it and do not repeat the retracted one.** This is a statement about **how difficult it is to construct a valid nonlinear instantiation of exact confounding**, not a statement that Theorem 1 fails once instantiated — since none of the summary-matched constructions were shown to be valid instantiations in the first place. The corrected, defensible claim is: matching a small number of chosen statistics between two mechanisms in a realistic market is not a reliable way to produce the kind of full observational equivalence Theorem 1 describes, and attempting it is easy to get wrong in ways that are hard to predict from the linear model alone.

What this means going forward. Theorem 1 remains an exact, useful diagnostic for the linear-Gaussian model it is proved for; nothing in Section 7 bears on that. What Section 7 does establish is a caution for anyone wanting to build a nonlinear analog of a linear identifiability result for validation

purposes: doing so correctly requires an explicit account of which statistics are matched, which features an observer can access, and how much of the incident’s aggregate footprint remains after matching — none of which the linear model specifies, and none of which we found a way to get right by intuition alone across our four *distinct-mechanism* attempts (the original, Study 5, and the third and fourth designs, all of which only ever matched a chosen summary statistic between genuinely different mechanisms). **One narrower construction does validly instantiate Theorem 1’s premise, and we should not claim otherwise:** Section 7.10.1’s negative control uses two labels with *identical* observable order-generation kernels, which trivially satisfies full observation-law equality by construction, not merely a matched summary statistic — and its near-chance result is exactly what Theorem 1 predicts for that case, consistent with (not surprising confirmation of) the theorem. The open, unresolved question is narrower than “whether any valid instantiation exists at all”: it is whether a nonlinear construction with two genuinely *different* underlying mechanisms (not merely differently-labeled identical behavior) can be engineered to validly satisfy Theorem 1’s premise — that is the question our four distinct-mechanism attempts did not resolve, not the question of whether identical-kernel equivalence is achievable in a nonlinear simulator at all, which the negative control already answers affirmatively and trivially.

7.4. A practitioner checklist

Grounded strictly in what the five studies (and their replications) showed, not extrapolated beyond the data — **revised after Section 7.5’s round-2 findings; items 2 and 3 are corrected below, not merely repeated:**

1. **Do not assume a linearized identifiability analysis’s premise is easy to instantiate correctly in a nonlinear deployment.** Across every attempted construction in this study, none was confirmed to actually satisfy Theorem 1’s premise (full observation-law equality) — engineering that equivalence correctly, rather than merely matching one or two convenient statistics, is itself a nontrivial and easy-to-get-wrong task.
2. **Feature richness matters, but the specific feature matters far more than richness per se.** Section 7.5 isolates signed executed volume as the single feature responsible for Study 4’s entire ranking

recovery; order-flow imbalance and queue concentration, added individually, recovered nothing. Don't chase "more features" — chase the specific microstructure quantity your feature set is missing.

3. **Footprint size, on its own, is not the deciding factor it first appeared to be.** Section 7.5's dedicated sweep found the reversal present across roughly a 15x range of incident footprint (1.4%–22% of volume) using the original feature set — footprint alone does not explain Study 5's different behavior. Don't assume a small-footprint incident is automatically the hard case; check the feature set first.
4. **Don't rely on a single classifier as a safe default without checking a stronger one alongside it.** The corrected Study 2 result shows a weak classifier can land on a different ranking than a stronger one on the *same* data — a single baseline's result is not conclusive evidence about the underlying identifiability structure.
5. **Replicate across seeds before trusting any single run's ranking.** The original Study 5 single-run read as "no signal anywhere"; five-seed replication revealed a real, if weak, signal on one side. A single calibration/simulation draw is not sufficient evidence either way.
6. **Population size and session length are lower priority to check than feature set.** Study 3 found the reversal stable across a 4x population range, and Section 7.5 found it stable across a wide footprint range too — feature choice, not scale, is where the real sensitivity lives.
7. **There is no theory-only way to know which regime you are actually in, and the dichotomy itself may not be stable.** Three independently-designed injection constructions (the original, Study 5, and Section 7.5's third design), plus one feature-engineering variant of the original (Study 4), produced four qualitatively different classification profiles. This is not merely "hard to predict in advance" — it suggests the confounded/separated labels do not have one fixed operational meaning across reasonable real-market constructions. See Section 7.6 for why three designs, not more, is treated as sufficient evidence for this claim.
8. **Investigate surprising results before either dismissing or trusting them — and note that "uninformative" is not the same as "low stakes."** Section 7.5's below-chance footprint result was checked with the same rigor as the Study 2 bug and confirmed genuine at the lowest footprint level tested, though a broader claim about a consistent below-chance pattern across low footprints did not survive a corrected,

independent-replication check and has been narrowed accordingly (see Section 7.5’s revised text). More importantly, Section 7.7’s cost extension shows a near-chance classifier (Study 5) is not a moderate-cost middle ground — its absolute cost substantially exceeds profiles with real discriminative signal (we withdraw an earlier “2–6x” ratio claim, which was arithmetically wrong given a near-zero denominator; see Section 7.7 for corrected absolute comparisons). Don’t assume every surprising number is an error, don’t assume every surprising number is real, don’t assume ten resplits of one dataset are ten independent checks, and don’t assume a “weak signal” result is a low-stakes one — check with genuinely independent data, and cost it out.

9. **Verify your simulator’s determinism claims independently of your own code.** Section 7.5 found a real reproducibility bug in ABIDES’s own RNG handling (`get_wake_time()` uses an unseeded global generator) that breaks bit-for-bit reproducibility across process runs despite this project’s own code seeding correctly throughout — “my code seeds everything” is not the same claim as “my results reproduce exactly.”
10. **Check whether your model injects the incident at the output or the state level before trusting any confounding claim.** Section 4.1’s Theorem 2 shows this distinction alone can flip an exactly-confounded pair into a highly classifiable one, with no nonlinearity required. **A precision, corrected here to match Section 4.1’s own qualification #1:** the operative distinction is *state-level vs. output-level* injection, not *persistent vs. one-time* — a one-time state-level perturbation is governed by the same unobservability condition as a persistent one (an earlier version of this item bundled “one-time” with “sensor-level,” conflating the two distinctions). A state-level disturbance, whether one-time or repeated, is generically far more fragile to confound than an output-level one, and this should be checked before assuming a theoretical confounding result transfers even to a linear operational setting, let alone a nonlinear one.

7.5. *A second round of validation: pinpointing the drivers, and one new reproducibility finding*

A second, more targeted round of experiments converts the qualitative claims above into precise, checked ones — and revises two of them.

Feature richness: one specific feature, not general richness, drives Study 4’s recovery. Testing each of Study 4’s three added features individually against the original 4-channel baseline isolates the effect precisely: adding *signed executed volume alone* reproduces the full ranking recovery (SEPARATED 1.000 accuracy vs. CONFOUNDED 0.829), while adding order-flow imbalance alone or queue concentration alone recovers nothing (both stay at baseline: SEPARATED ≈ 0.72 , still below CONFOUNDED’s ≈ 0.85). This revises Section 7.3’s framing: “richer features” was imprecise. What matters is whether the observer’s feature set captures signed, executed (not merely resting) order flow — a specific microstructure quantity, not feature-set size in general.

Footprint size does not explain the reversal — it was the wrong hypothesis. A dedicated sweep, varying the incident’s aggregate order-flow footprint from roughly 1.4% to over 20% of total market volume using the *original* 4-channel feature set, found the reversal (CONFOUNDED > SEPARATED) present at essentially every footprint level tested — e.g. at a $\sim 10\%$ footprint, SEPARATED accuracy is 0.567 against CONFOUNDED’s 0.733; at a $\sim 22\%$ footprint, 0.867 against 0.933. This corrects Section 7.3’s earlier attribution of Study 5’s muted signal to footprint size: footprint alone, holding the feature set at baseline, does not turn a reversal into a recovery or vice versa. Whatever distinguished Study 5’s design must trace to some other construction choice (attribute subset, calibration target, or clock structure); we do not have a clean isolated explanation for it beyond what Section 7.2 already reported.

At the very lowest footprint tested ($\sim 1.4\%$), SEPARATED accuracy fell *below* chance (0.367). Investigated with the same scrutiny applied to the Study 2 threshold-rule bug: this is **not** a hardcoded-sign artifact (this classifier learns its orientation from training data), and an initial resampling across 10 additional train/test splits of the *same* dataset found 8/10 below chance. **We have since found this specific replication method to be inadequate and are correcting the claim rather than repeating it:** 10 resplits of one fixed dataset are not 10 independent experiments, since they share the same underlying episode pool. A corrected check using genuinely independent data realizations at two footprint levels found the pattern does not hold uniformly — 8/10 independent replicates below chance at the $\sim 1.4\%$ footprint level, but only 5/10 (exactly consistent with chance) at a $\sim 3.5\%$ footprint level, where the original same-dataset-resplit method had suggested 9/10. **We now read the lowest-footprint result as more**

likely genuine but no longer treat the broader “consistently below chance across low footprints” pattern as established — it held at the very lowest level tested under independent replication, but did not extend to a nearby, only moderately higher footprint level the way the original (inadequate) method suggested.

A fourth, distinct outcome from a third independent injection design. A third independently-designed incident agent (built from the attribute specification only, varying a different attribute subset and calibration target than either the original design or Study 5) produced yet another distinct pattern: SEPARATED reached near-perfect accuracy using only the *original* 4-channel feature set, while CONFOUNDED simultaneously reached a high (though not near-perfect) level, despite the tightest order-count calibration match yet achieved (3.3%, versus roughly 2x in the original design).

Both figures have since been regenerated at large N ($N = 1000/\text{class}$) under the fully-hardened pipeline, after this design’s own `config_builder_v3.py` was found never to have been patched for the RNG bugs at any point in the project — and the two regimes behaved differently under that correction, which is itself the informative result. The separated regime held up essentially unchanged (0.989/1.000 pre-patch $\rightarrow$ 0.985/0.999 hardened), **ruling out the RNG bug as an explanation for this design’s near-perfect separated-regime performance** — that result is genuine. The confounded regime did *not* hold up as well: it fell from 0.889/0.969 to **0.845/0.911** (-4.4pp accuracy, -5.9pp AUC), meaning **part of this design’s apparent confounded-regime near-separability was an artifact of the unpatched RNG, not a property of the construction.** We report both directions plainly: one claim survived a genuine test, the other was partly inflated by a bug and is corrected downward here. The qualitative finding (this design discriminates well in *both* regimes, unlike the others) survives, but its confounded-regime magnitude was overstated in earlier versions.

The mechanism is directly diagnosable: this design’s compromise and fault mechanisms differ sharply in `cancel_rate`, which is directly one of the four observed channels, so matching total order count left cancellation behavior free to leak a strong signal. **To be precise about the count: three independently-designed injection layers exist (the original, Study 5, and this design), which together with one feature-engineering variant of the original (Study 4) produce four qualitatively different classification profiles:** reversed (original, baseline features), fully recover-

able via one specific feature (original design, expanded features — Study 4), a muted middle ground (Study 5), and simultaneously strong-on-both-regimes via a different leaking channel (this third design). We read this as direct evidence that the confounded/separated dichotomy does not have one fixed operational meaning in this class of nonlinear market — which channel leaks, and how much, depends on exactly which attributes a designer chooses to vary versus match, in a way the linear model’s fixed, given (C, b_K, b_F) never has to confront.

A separate reproducibility finding, unrelated to the substantive results. While investigating the below-chance footprint result, attempting to regenerate it revealed that identical seeds do not produce bit-identical episodes across separate process invocations of this codebase, traced to ABIDES’s own `get_wake_time()` utility (scheduling most of the background population’s wake-up times) calling Python’s unseeded global random number generator rather than the codebase’s own correctly-seeded random state objects. This is a bug in ABIDES’s third-party code, not in any script written for this project. **This bug, along with two further unseeded-RNG bugs found during the same investigation, was subsequently patched (Section 7.10.5)** — see that section and the Appendix for the corrected reproducibility status; at the time this specific below-chance result was generated, the fix had not yet been applied, so this particular result and everything in Sections 7.1–7.8 should be read with that caveat, distinct from the `str_to_ns` bug (Appendix), which blocked correctness outright and had to be fixed before any ABIDES results could be trusted at all, versus the RNG bugs, which affect only bit-for-bit reproducibility of specific seeds, not the validity of any individual run’s statistics.

A note on the broader recognition of this transfer question. Section 2 engages the closest analog to this finding — Alisic & Amin’s (2026) geometric proof that sensor-attack stealthiness does not automatically transfer from linear to nonlinear (Lie-group) operating conditions — in detail, including an open question about whether their commutativity-style condition could be adapted to predict, rather than only explain after the fact, the reversal reported here. We do not repeat that discussion here; the point relevant to this section is that the linear-to-nonlinear transfer question is a live, independently-recognized concern in the adjacent literature, not an artifact of this paper’s particular framing.

7.6. *Why three injection designs, and not more: a stopping-rule justification*

Three independently-designed injection layers is a deliberate stopping point, not an arbitrary one, and not merely a budget constraint. The justification: Section 1 names five controllable dimensions of an ABIDES incident (directional bias, size, rate, price aggressiveness, cancellation churn) — these are the concrete, simulator-specific instantiations of the abstract injection direction b_M in Section 3.2’s linear model, not a table given in Section 3.2 itself, which states the general model only. The three designs tested differ in exactly the way that matters for this paper’s argument — which subset of those five dimensions is varied versus held fixed during calibration, and against which target statistic calibration is performed. The original design varies all five and calibrates against mean price drift; Study 5 varies a different subset with a different (spread-fraction, deterministic-clock) parameterization and a different calibration target; the third design varies yet another subset and leaves `cancel_rate` uncalibrated, which is precisely what produces its distinctive leakage. Between them, the three designs span the qualitatively distinct calibration failure modes identified across this section: leakage through activity-level mismatch (original), an as-yet-unexplained muted signal (Study 5 — **we no longer attribute this to footprint size**, since Section 7.5’s dedicated sweep later found footprint alone does not explain it; that attribution is retracted here rather than repeated), and leakage through an uncalibrated discrete-behavior channel (third design). We do not claim these three exhaust the design space — **a fourth design was subsequently built specifically to test this, and did surface a new outcome, reported in Section 7.9 rather than duplicated here**. What we do claim is that three independently-motivated designs, chosen to differ along the dimensions Section 1 itself identifies as the relevant attack surface, already established the paper’s core claim (the dichotomy is construction-dependent, not fixed) with enough distinct failure modes to rule out “we got unlucky once” as an explanation — a claim about instability, not about a complete taxonomy of that instability, which remains future work.

A matching-tolerance audit, per external review, sharpens what “matched” means here and should be read alongside the above. No pre-specified numerical tolerance was set before this project began measuring how close each design’s calibration search actually got to its target statistic. Auditing all four designs after the fact, on independent (non-calibration) data, against even a generous tolerance ($|\text{Cohen’s } d| < 0.1$ on the target statistic): **zero of four designs achieve it**, including the original design

(closest, $d = -0.129$), Study 5 ($d = -0.210$), the third design ($d = 0.527$), and the fourth design (worst, $d = 0.933$). **Every instance of “matched” or “summary-matched” in this section should be read as “match attempted, not match achieved to any pre-specified standard”** — a materially more honest framing than implying a successful match followed by unexpected leakage. **We attempted to close this gap for the closest design (V1, $d = -0.129$) with a finer, 16-point calibration grid searched against independent held-out data at every point. A precision worth being exact about, per external review: the resulting apparent match ($d = 0.003$ at $\theta = 1.8$) came from *selecting* the best-looking candidate among 16 grid points — this selection step means the apparent match itself is not an untouched confirmatory result, even though the grid’s underlying episodes were independently generated.** Two subsequent, genuinely fresh confirmation runs at this *same selected* $\theta = 1.8$ (not re-running the selection) then failed to reproduce it: $d = -0.572$ on the first re-run, and $d = -0.445$ on a second, larger ($N = 150/\text{class}$), independent confirmation. **The correct, precisely-scoped claim: at the selected value $\theta = 1.8$, two fresh confirmation runs failed the target tolerance.** We withdraw the earlier framing of this as “three independent tests” confirming a “structural property of the calibration problem,” which overstated what a two-fresh-confirmations-of-one-selected-setting result actually shows. **This has since been settled definitively, closing the “systematic mismatch vs. pure noise” question the previous version left open.** Ten further independent fresh batches were run at $\theta = 1.8$ ($N = 150/\text{class}$ each), giving 12 independent d -estimates in total (2 prior confirmations + 10 new): mean $d = -0.4643$, SD 0.0736, 95% CI $[-0.511, -0.418]$, $t(11) = -21.84$, two-sided $p \approx 2 \times 10^{-10}$. **All 12 estimates are negative, tightly clustered (range $[-0.572, -0.350]$), far from the original grid search’s single apparent “hit” of $d = 0.003$.** This is a systematic, statistically robust non-zero effect, not noise around zero: $\theta = 1.8$ consistently overshoots, making F ’s mean drift substantially exceed K ’s — the opposite of the intended match — and the original grid search’s apparent success is now conclusively a small-sample fluke, not a real calibration point. This does not establish that no fixed θ can meet tolerance for this design, or that the result generalizes to the other three designs, but it does close the specific question of whether $\theta = 1.8$ ’s failure was noise (it is not) — and we did not pursue the same exercise for the other three (already worse-matched) designs. This does not undermine the paper’s core argument

(that engineering a valid nonlinear analog of exact confounding is difficult) — the definitively-settled claim (the one selected calibration point for the closest design fails tolerance systematically, not by chance) is, if anything, a stronger version of that argument than what we previously claimed.

7.7. Cost tables: fully recomputed with the positive fault-loss correction from Section 6.3 (illustrative dollar-scale reconstruction, not an empirical calibration — Section 6.2’s caveat applies identically here)

This table has been fully recomputed, not relabeled. An earlier version of this section corrected the prior-shift threshold bug but retained the original zero-fault-loss assumption ($H_F = 0$) throughout — meaning every number below was, despite surrounding text describing a “correction,” still computed under the assumption Section 6.3 explicitly retracted. This was caught by external review via a direct arithmetic check (the old table’s own “New CoA” minus “New excess vs. best-constant” columns reproduced the exact \$154.85K–154.9K *zero-fault-loss regret in every row, regardless of design*) and *discorrected* the $C_F = \$473,958.33$, $H_K = \$1,317,500$) together with Table 2c’s directly-measured, design-and-regime-specific H_F values, for five profiles (four independently-designed constructions plus Study 4, the feature-engineering variant of the original design) $\times$ 2 regimes $\times$ 5 priors (50 cells total). **This corrected table’s own implied baselines have since been independently checked against the exact accounting identity $B(p) = \min\{p\Delta_K, (1-p)\Delta_F\}$ by external review and confirmed consistent to within rounding across all 8 original rows — the specific 154.85K-vs-144.4K discrepancy that motivated this recomputation is closed.**

A coverage gap from an earlier version of this recomputation, now closed: Study 4 was inadvertently dropped from this table when V4 was added, and has since been restored. Study 4 shares its underlying incident mechanism bit-for-bit with the original design (V1) — confirmed directly by comparing `action_log` output under identical seed/mechanism/direction/magnitude, which matched exactly, including total order count and shares — so it correctly reuses V1’s ground-truth-measured H_F rather than requiring independent measurement; only its feature set (21 channels vs. the original’s 4) differs.

Table 4 (fully recomputed, now including Study 4, and now using direct class-conditional weighting rather than resampling — see below). Mean-across-5-priors summary, by design/profile and regime.

Design/profile	Regime	H_F used	Mean regret vs. oracle	Mean excess vs. best-constant	Mean FPR	Mean FNR
Original (V1)	Separated	\$33.09K	\$121.9K	−\$29.7K	0.169	0.623
Original (V1)	Confounded	\$104.67K	\$53.6K	−\$90.8K	0.096	0.264
Study 4 (ex-panded features)	Separated	\$33.09K	\$3.6K	−\$148.0K	0.000	0.019
Study 4 (ex-panded features)	Confounded	\$104.67K	\$50.6K	−\$93.7K	0.054	0.313
Study 5 (V2)	Separated	\$0.85K	\$156.5K	+\$1.7K	0.262	0.702
Study 5 (V2)	Confounded	\$0.85K	\$161.3K	+\$6.5K	0.227	0.747
Third design (V3)	Separated	\$1.81K	\$7.8K	−\$146.9K	0.013	0.058
Third design (V3)	Confounded	\$4.91K	\$43.8K	−\$110.6K	0.076	0.200
Fourth design (V4)	Separated	\$5.85K	\$130.6K	−\$23.6K	0.173	0.636
Fourth design (V4)	Confounded	\$41.24K	\$0.0K	−\$150.7K	0.000	0.000

A methodological upgrade, per external review: the single-resampling-draw cost estimate used in earlier versions of this table has been replaced with a direct class-conditional weighting formula, $\hat{R}_p = p\{C_K + (H_K - C_K)\hat{\text{FNR}}\} + (1 - p)\{H_F + (C_F - H_F)\hat{\text{FPR}}\}$, computed

once per cell from the held-out test set’s FPR/FNR at the fixed cost-optimal threshold — removing the resampling-draw noise entirely, since this formula has no random resampling step. This is not a cosmetic change: comparing the two methods across all 50 cells, the maximum absolute difference is \$8,561 (2.02%, at the fourth design’s separated regime, $P(K) = 0.5$) and the mean absolute difference is \$1,893 — small relative to the overall cost scale, but large enough to change which specific cells are flagged below. The table above uses the corrected, direct-weighting values throughout, superseding the resampling-based figures reported in earlier versions of this section.

Study 4’s richer feature set produces a substantially larger cost advantage than the original 4-channel set — separated-regime regret drops from \$121.9K to \$3.6K, and excess-vs-best-constant improves from $-\$29.7\text{K}$ to $-\$148.0\text{K}$ (figures now matching the direct-weighting table above; an earlier version of this sentence quoted the superseded resampling-based values) — consistent with, and now quantifying in dollar terms, Section 7.2’s finding that the expanded feature set recovers much of the theoretical ranking. Study 5 remains the one profile with no measurable classifier benefit over the best constant policy (small positive excess in both regimes); the other four profiles’ classifiers substantially beat the best constant.

A distinction the earlier version conflated, now reported explicitly and made fully auditable per external review: “beats always-contain” and “beats best-constant” are different claims. An earlier version stated an unverifiable count (“7 of 40 cells”) from the 8-row averages table alone; **we have since built the complete 50-cell per-prior table with an explicit, stated criterion** (point-estimate comparison, strict inequality — not a statistical significance test, stated plainly so the count is well-defined and auditable), using the identical classifier (LogisticRegression) and train/test split across all five profiles. **The count itself has since been refined once more, per external review, by replacing single-draw resampling with direct class-conditional weighting** (Section 7.7’s methodological upgrade, above) — the two methods disagree on which specific cells cross the boundary, so we report the corrected count rather than the resampling-based one: **50 of 50 cells beat always-contain; 41 of 50 beat best-constant; 9 of 50 beat always-contain but not best-constant:** original-design-separated at $P(K)=0.05, 0.10$; V2/Study 5-separated at $P(K)=0.05, 0.10, 0.20$; V2/Study 5-confounded at $P(K)=0.05, 0.10, 0.20$; V4-separated at $P(K)=0.05$.

All nine are low-prior cells in which always-leave is the best constant policy, so the classifier’s failure to beat it is a failure to add value when the base rate is already low, not a failure to beat naive containment. (An earlier count on the pre-unification data read 48/42/6 and reported two Study 5 cells at $P(K)=0.50$ failing to beat even always-contain; that reading does not survive on the unified large-N basis and is withdrawn.) The full 50-cell table on the unified basis is provided as supplementary data (`table4_full_50cell.csv`, generated from `task5_final_unified_cost_table.csv`).

What this means for the earlier three-profile narrative (Study 5 evaporates, original contrast reverses to benefit, Study 4/third design unaffected): the qualitative story survives this recomputation across all five profiles now covered — Study 5 is still the profile with no real cost benefit (and, per the full table, is also the only profile with cells failing to beat even the weakest baseline), and the other four profiles still show a genuine classifier advantage — but the specific dollar figures reported for that narrative used the wrong loss model and should be read as superseded by the table above.

The scope decision, updated: every profile reported in Section 7 now has a re-verified, prior-corrected, positive-fault-loss dollar figure, auditable at the individual-cell level — closing the coverage gaps (V4, then Study 4, both previously missing), the correctness gap (the prior-shift threshold bug), the loss-model gap (the zero-fault-loss assumption Section 6.3 retracted), and the auditability gap (the full 50-cell table, not just an 8-row summary) in one pass. None of these figures should be described as “calibrated” in any submission, per Section 6.2’s correction, which applies identically here.

7.8. *A persistence/duration ablation (reframed — this is not a clean causal test of Theorem 2)*

Section 4.1’s Theorem 2 predicts that a *persistent* (state-feedback) disturbance is far more fragile to confound than an equivalent disturbance that does not compound. We attempted to test this directly by holding feature set and nominal footprint parameters fixed and varying only whether the incident’s behavioral shift persists across wakeups or fires once and reverts to baseline. **On external review, this comparison does not isolate the mechanism it was designed to test, and we correct the interpretation here rather than let the stronger causal claim stand.** A one-off disturbance still changes the order book and can trigger downstream market reactions — a one-off order is not equivalent to “no state feedback,” only

to a much shorter and less repeated one. The comparison simultaneously varies repeated forcing, incident duration, cumulative order activity, and the amount of evidence generated, not persistence alone; equal *parameter settings* between the two conditions do not establish equal *realized* footprint (order counts, executed volume, active duration were not measured and matched directly). Lower classifiability from a much shorter, non-repeated incident is an expected finding on general grounds, whether or not state feedback specifically contributes anything.

What this result actually supports, stated at the correct strength: shorter, non-repeated incidents were less classifiable than persistent ones in these specific experiments — a genuine, useful duration/persistence finding in its own right, and directionally consistent with (though not a clean isolated confirmation of) Theorem 2’s prediction. It does **not**, on its own, establish that state-feedback fragility specifically (as opposed to reduced duration, reduced cumulative activity, or reduced evidence generation generally) explains a specific, quantified share of Section 7’s earlier findings. A design that would isolate the mechanism — comparing reactive against frozen/replayed background market behavior under matched incident order schedules (same commanded order counts, submitted volume, and active duration in both conditions), with dose (single vs. repeated forcing) varied as a separate factor — **has since been built and run (Section 7.10.2); an earlier version of this sentence said it had not been, which is now stale.** See Section 7.10.2 for the (inconclusive) result.

Original numbers, retained with the corrected interpretation above (and with one further caveat added on reflection): removing persistence reduced classifiability in both scenarios (10-split robust means): the contrast scenario fell from 0.695 to 0.530 accuracy (0.733 to 0.559 AUC); the summary-matched scenario fell from 0.829 to 0.478 accuracy (0.921 to 0.472 AUC) — the latter reaching a level consistent with chance. **The 0.829 baseline used here predates the large-N reverification reported in Section 7.1, which found the summary-matched regime’s true accuracy is closer to 0.89** — this ablation’s own before/after comparison has not been re-run at that corrected baseline, so the relative *size* of the persistence effect on the summary-matched side should be treated as provisional pending that check, even though the qualitative finding (removing persistence substantially reduces classifiability) is unlikely to be affected in direction. This is a real, large, and reportable duration/persistence effect; it is not, as an earlier draft claimed, a clean separable confirmation that “state-

feedback fragility is doing real explanatory work,” since duration and dose were not isolated from persistence in this design.

An additional observation, not smoothed over: the ranking between the two scenarios’s one-off accuracies (0.530 vs. 0.478) is technically in the same direction as the original pilot’s ranking, but only marginally — both values sit close to chance rather than showing a clean, large gap. Given this section’s reframing, we do not read this as “restoring Theorem 1’s prediction” (since neither scenario was shown to instantiate Theorem 1’s premise in the first place) — we simply note that both scenarios’ classifiability dropped substantially and roughly comparably when persistence was removed, consistent with the duration/dose confound identified above rather than pointing to any specific, isolated mechanism.

7.9. A fourth design: exploratory extension, not a confirmatory test

A framing correction here, per external review: an earlier heading called this “confirmation” of a stated prediction — this overstates what an exploratory extension can establish. Section 7.6 predicted, in general terms, that a fourth design would “very plausibly” surface a new outcome; it did, but a vague directional prediction being borne out by continued exploration is not the same epistemic event as a pre-registered, specific prediction surviving a genuine test. We report the fourth design’s result as what it is — one more data point in an ongoing exploratory process, consistent with Section 7.6’s own explicit disclaimer of exhaustiveness — not as confirmatory evidence in the stronger sense that phrase can imply.

Section 7.6 explicitly declined to claim the three tested injection designs exhaust the space of possible outcomes, stating plainly: “*a fourth or fifth design would very plausibly surface a fifth or sixth specific outcome.*” We tested this prediction directly by building a fourth independently-designed incident agent (attribute subset {`buy_bias`, `size_mult`, `aggressiveness`}, calibrated against mean spread — a combination distinct from all three prior designs) and found **exactly what Section 7.6 predicted:** a fifth qualitatively distinct outcome, not a repeat of any existing pattern.

Table 5. All five outcomes, side by side. All figures now re-generated at large N ($N = 1000/\text{class}$) under the fully-hardened pipeline (see Section 7.10.9); pre-patch values shown where they differ meaningfully.

Design	SEPARATED acc/AUC	CONFOUNDED acc/AUC	Reversed?
Original	0.668 / 0.743 (unified basis; Section 7.1's large-N recheck reported 0.695 / 0.733 — see note below)	0.892 / 0.954 (up from 0.829 small-sample — see Section 7.1)	Yes
Study 4 (feature variant)	0.995 / 1.000 (was 0.991/1.000; no meaningful change)	0.870 / 0.938 (was 0.886/0.933; no meaningful change)	No
Study 5	0.615 / 0.661 (was 0.522/0.588 pre-patch — +9.3pp acc, +7.3pp AUC under hardening)	0.558 / 0.579 (was 0.544/0.547 — +3.2pp AUC)	No (both weak, but less so than first reported)
Third design	0.985 / 0.999 (was 0.989/1.000; no meaningful change — near-perfect separation confirmed genuine, not an RNG artifact)	0.845 / 0.911 (was 0.889/0.969 — −4.4pp acc, −5.9pp AUC; part of the apparent near- separability was an RNG artifact)	No
Fourth design	0.693 / 0.759 (was 0.644 / 0.769 small-N)	0.998 / 1.000 (Section 7.9's zero-error run: 1.000 / 1.000 — see note below)	Yes — the most extreme reversal of any design

Note on the unified basis (r18). The original and fourth designs were regenerated under the verified-clean pipeline using the exact seeds of their earlier runs and produced bit-identical feature arrays — establishing that their earlier data was never contaminated. The values above are then all evaluated on one common basis (seed-1 70/30 split, 600 held-out episodes per profile) so that the five rows are directly comparable. Two earlier figures do not match this basis exactly and have not yet been reconciled: the original design’s contrast-regime 0.695 / 0.733 (Sections 7.1, 7.8) and the fourth design’s zero-error confounded run (Section 7.9, Appendix). Both differences are small and neither changes any qualitative claim, but the source of each (evaluation split, seed block, or pool size) must be identified and stated before submission rather than left as two coexisting numbers.

The fourth design reverses in the same direction as the original but far more extremely: its confounded case reaches *perfect* classification, the highest of any design tested. **This specific cell was re-verified at large N (1000 episodes/class, 2000 total, under the verified-clean pipeline after fixing two previously-unwired RNG bugs in this design’s own config file) following the contrast-regime instability discovered elsewhere in this section — unlike that instability, this result is not a small-sample artifact.** We correct our own initial statistical treatment here, per external review: a bootstrap interval on a zero-error test set is degenerate (it can only resample errors that were observed, and none were) and cannot legitimately be reported as “essentially zero width.” **The correct interval is the exact Clopper-Pearson binomial bound given zero observed errors on the held-out set: [0.9939, 1.0], using the correct denominator of 600 held-out test episodes (2,000 total episode pool, 1,400 training, 600 held-out — counts confirmed and reported explicitly per external review, since the interval’s validity depends on using the test-set size, not the full pool) — still a very tight, high-confidence result, but a genuine interval with real width, not the vacuous one-point interval we originally reported.** A 30-split sanity check gives mean accuracy 0.999 (SD 0.0014) across every partition of the 2000-episode pool, consistent with (not a substitute for) the exact interval above. This cell is genuinely, and defensibly, saturated. The mechanism is again directly diagnosable and again a variant of the same underlying pattern: calibrating against one statistic (mean spread) left a large, uncontrolled asymmetry in a different attribute (effective order size, 5.5x vs. 1.5x baseline between the two mechanisms) that leaks strongly through depth-related features (1.76–1.88

pooled-standard-deviation gaps, dwarfing the calibrated spread gap of 0.85). **This is the fourth independent replication of the “confounding by search always leaves something to leak” pattern first identified in Section 7.5** — now via a size-based leak rather than the cancellation-rate leak of the third design, confirming the pattern is not tied to any single attribute.

What this means for Section 7.6’s stopping rule, stated at the corrected strength: this is consistent with, not confirmatory proof of, the original justification — Section 7.6 predicted a fourth design would plausibly surface something new, in general terms, and this exploratory extension did. The instability observation is now based on five outcomes across four independent designs, with no sign of convergence toward a single stable pattern; the complete characterization of that instability remains explicitly open, as Section 7.6 always framed it, and we do not treat this fourth design as having tested or confirmed anything beyond what continued exploration would be expected to find.

7.10. Response to external review: a negative control, a corrected cost pipeline, and closing eight identified gaps

Following the mathematical and methodological corrections in Sections 4, 6, and this section’s opening reframing, eight further gaps identified by external review were addressed directly, with genuinely mixed results — clean validations alongside a materially worse finding and one still-open complication. We report all eight plainly.

7.10.1 Negative control — a genuine, if partial, pass, after a real diagnostic journey and a statistical correction to how “decisive” was measured. Two incident-agent variants with *identical* observable order-generation kernels (same distribution over rate, size, side, aggressiveness, and cancel behavior; differing only in an internal, behaviorally-inert label) were run through the full classification pipeline. The initial check (5 seeds) gave mean accuracy 0.533 (range 0.333–0.700) — consistent with chance, but too weak to rule out subtler concerns. A strengthened version (bit-identical exogenous-stream coupling confirmed working correctly; 10 reps) instead gave a mean of **0.43** — **noticeably below chance**, not a clean confirmation. Investigating rather than re-running blindly: no orientation or sign-convention bug was found in the classifier or label assignment, but **K and F episodes were being drawn from two fixed, disjoint random-seed blocks** — a controlled test (real labels vs. randomly-shuffled labels

on the same 500 episodes) confirmed this was producing a genuine, if small, systematic effect. Fixed by interleaving K/F seed assignment from one shared pool rather than disjoint blocks.

A large-N, interleaved-seed check was then run and initially reported using 20×5 -fold cross-validation, which we withdraw as the wrong tool here, per external review. Repeated cross-validation on one fixed dataset measures variance across partitions of the *same* episodes, not uncertainty over new, independent episodes — folds and repeats share data, so the resulting interval (accuracy/AUC both $[0.492, 0.498]$) understates true uncertainty and should not have been called decisive. **We replaced it with the statistically correct comparison: 20 genuinely independent replicate datasets** (fresh episodes each time — 100 episodes per replicate, 2,000 total; confirmed by direct source inspection that each replicate’s K and F halves are drawn from disjoint seed pools with no shared exogenous stream, so no pairing structure exists that could be broken by the standard unpaired train/test split used here), with a between-replicate confidence interval computed across those 20 replicate-level values rather than within one of them.

We now have the exact interval bounds, computed two ways, and report both plainly rather than a rounded summary. Mean accuracy across the 20 replicates is 0.505 (SD 0.090); mean AUC is 0.483 (SD 0.089). By normal approximation: accuracy 95% CI $[0.466, 0.544]$, AUC 95% CI $[0.444, 0.522]$. By percentile bootstrap (10,000 resamples over the 20 replicate values): accuracy CI $[0.467, 0.543]$, AUC CI $[0.446, 0.521]$ — the two methods agree closely. **Checked against the pre-specified equivalence margin $[0.45, 0.55]$: accuracy’s interval is fully contained within the margin (a genuine equivalence pass); AUC’s interval is not** — its lower bound (≈ 0.444 – 0.446) falls just outside 0.45. This is a more precise and more honest verdict than our own earlier “point-estimate-level pass, interval unknown” hedge: accuracy passes a proper equivalence test; AUC’s point estimate is inside the margin but its interval is not, so AUC does not pass an equivalence test at this margin and sample size. The valid between-replicate interval width (accuracy: $0.544 - 0.466 = 0.078$) is **13 times** wider than the invalid within-dataset CV interval’s width ($0.492 - 0.498 = 0.006$) it supersedes ($0.078/0.006 = 13$) — matching our own earlier rough “12–13x” estimate exactly; **we correct an intermediate arithmetic error here that stated this as “roughly 8x,”** which was simply a computational mistake, not a refinement. We also note, per external review, a further dis-

inction worth being precise about: this 95%-interval-containment check is a conservative basis for equivalence, distinct from a formal two-one-sided-tests (TOST) equivalence test at $\alpha = 0.05$, which would use a 90% interval instead — an independent check using a one-sample t -procedure on the reported summary statistics (accuracy 90% interval $[0.470, 0.540]$, AUC 90% interval $[0.449, 0.517]$) gives the same accuracy-passes/AUC-does-not verdict under either framing, but the two are not interchangeable in general and we do not claim to have run a formal TOST here. **We report this as: accuracy is rigorously established as equivalent to chance at this margin; AUC is not, and the small residual gap (mean 0.483, not landing symmetrically around 0.5) has not been further diagnosed.** This finding concerns the mean performance of this specific fitted procedure at this sample size, not a claim about equality of the full observation laws or a bound applicable to every possible classifier.

7.10.2 Dose/persistence isolation — inconclusive, reported as such. The redesigned reactive-vs-replayed comparison (Section 7.8’s follow-up, isolating state feedback from duration/dose) was built and run, catching and fixing two further implementation bugs (an RNG-desynchronization issue and a latency-double-counting issue) along the way. **The 4-cell result at $N = 30$ per cell is genuinely inconclusive** — we do not have a large enough sample to distinguish a real feedback effect from noise in this design, and we report this as an open question rather than force a conclusion. Section 7.8’s duration/dose-confound correction stands; this experiment does not yet resolve it either way.

7.10.3 Frozen confirmation set — status corrected, this claim is superseded by 7.10.5. An earlier version of this subsection described this set as “clean, and the strongest validation available.” **That description is withdrawn, not merely qualified:** Section 7.10.5 found this specific set was generated under an internally mixed pre-/post-patch provenance, not the uniform post-fix status originally claimed. A genuinely verified-clean regeneration has since been produced (Section 7.10.5); this original 7.10.3 entry is retained here only as a historical record of what was claimed and later corrected, not as evidence.

7.10.4 Corrected cost pipeline — re-verified and now fully re-computed with the correct loss model, not just the correct threshold. This item combines two separate corrections that were previously conflated: the prior-shift threshold fix (Section 6’s bug) and the positive-fault-loss fix (Section 6.3’s audit). **Both are now applied together in**

Section 7.7’s fully recomputed table. The decomposition formula must also be updated to match, per external review — an earlier version of this and Section 7.10.8 still stated the zero- H_F identity $R(\pi) - R_{\text{oracle}} = p(\text{HARM} - C_{OP}) \cdot \text{FNR} + (1 - p)\text{DISRUPT} \cdot \text{FPR}$ as exact even after Table 4 moved to $H_F > 0$. The correct identity for the positive-fault-loss model used in Table 4 is:

$$R_{\text{oracle}} = p C_K + (1-p) H_F, \quad R(\pi) - R_{\text{oracle}} = p (H_K - C_K) \cdot \text{FNR} + (1-p) (C_F - H_F) \cdot \text{FPR},$$

with cost-optimal threshold $q^* = \frac{C_F - H_F}{(H_K - C_K) + (C_F - H_F)}$ — the zero- H_F formulas in Section 6 remain correct *for that section’s own toy scenario*, where $H_F = 0$ by construction, but should not be read as governing Section 7’s positive-fault-loss model. This distinction was not stated clearly enough in earlier versions and is corrected here. Under the complete correction: Study 5 remains the one profile with no measurable classifier benefit over the best constant policy (small positive excess in both regimes, and 2 of its 10 individual cells failing to beat even always-contain); the other four profiles’ classifiers substantially beat the best constant. Study 4 and the third design’s *qualitative* advantage survived the threshold correction, though — per a full per-episode check — only one specific cell (Study 4’s contrast regime) was confirmed literally unchanged at the individual-decision level; the other cells in these two profiles shifted modestly in magnitude while keeping the same sign. **Study 4’s own cost row, previously missing from Table 4, has since been restored (Section 7.7)** — an earlier version of this paragraph still flagged that gap as open; it is closed.

A further nuance, not to be smoothed over: the prior-shift fix is not a pure improvement — it trades one error type for another. Completing the false-negative-rate columns that a previous version of this section omitted (reporting only false-positive rate) reveals that the fix which lowers false-positive rates across every profile *raises* false-negative rates substantially — for example, Study 5’s contrast regime: FPR drops from 0.952 to 0.263, but FNR rises from 0.070 to 0.701. This makes sense mechanically (a threshold correctly adjusted for a low deployment prior becomes much more conservative about calling “compromise,” which necessarily lets more true compromises through uncaught) but means the corrected pipeline should not be described as simply “more accurate” — it is more accurate about *cost*,

given the specific cost triple used, but the error profile has shifted, not uniformly shrunk, and a deployment with a different cost triple (or a genuine preference for erring toward containment) would need to re-derive its own threshold rather than assume the correction is cost-triple-independent.

A separate methodological subtlety surfaced while re-deriving the FPR/FNR tables: the cost-decomposition identity $R(\pi) - R_{\text{oracle}} = p(\text{HARM} - C_{OP}) \cdot \text{FNR} + (1 - p)\text{DISRUPT} \cdot \text{FPR}$ holds exactly (to floating-point precision) when checked against the *realized* resampled prior in a given table row, but checking it against the *nominal* target prior p_k instead produces spurious apparent violations of up to 0.51 cost units — a reminder that any reader attempting to independently verify this paper’s cost tables should reconstruct the realized class balance from the raw predictions, not assume the nominal prior was achieved exactly in every finite resample.

7.10.5 RNG audit — three bugs total, two found in this round, plus a fourth, subtler provenance bug found on further investigation, and a fifth found on top of that. Beyond the originally-disclosed `get_wake_time()` issue, two further unseeded-RNG bugs were found and patched. Bit-identical cross-process reproducibility was demonstrated only after all three fixes were applied. **A subsequent investigation then found that our own patch-verification process was itself unreliable:** worker processes performed their own lazy, per-worker import of the affected modules and cached that import for their entire lifetime, so results generated *after* the patch was written to disk could still reflect pre-patch code if the worker that produced them had already cached the old version — a subtler failure mode than simple parent-process staleness, confirmed by direct reproduction and fixed with a forced fresh re-import plus a behavioral fingerprint check (`verify_patch_active()`) that verifies all three patches are actually active in a given worker, independent of file timestamps. **External review correctly identified that even this fingerprint-based fix was not sufficient on its own:** reimporting a module does not update references already held by objects or workers created before the reimport, and a further investigation confirmed this concern was valid and worse than described — the fingerprint check itself gave a false positive in one deliberately-constructed test (following a real mid-run file reversion), and a second failure mode was found where a re-exporting parent package silently defeated what looked like a “fresh” submodule import. **Both are now closed** by killing and respawning every worker process fresh per task (`maxtasksperchild=1`), verified to eliminate the gap completely *for the specific caching failure mode diagnosed*.

A further precision, per external review: `maxtasksperchild` controls worker recycling, not the multiprocessing start method (fork vs. spawn) — a forked worker can still inherit a stale parent’s already-imported state even when recycled per-task, whereas `spawn` creates a genuinely fresh interpreter. A full project-wide audit of every production script’s actual start method has since been completed, closing this gap: 9 scripts (an early era) use plain `fork` with no hardening at all; 3 scripts (a middle era) use `spawn` but without `maxtasksperchild`; only scripts written from the point this fingerprint fix was introduced onward use the fully-hardened combination (`spawn + maxtasksperchild=1`). **For every retained result in this paper, the vulnerability this section diagnoses is confirmed theoretical, not actual:** the early-era scripts’ results either predate the original three-bug patch entirely, or used a config-builder variant that was never patched at all (a documented fact, not a timing coincidence); the middle-era scripts’ results were confirmed via direct file-modification-time inspection to have run entirely within a stable, unedited-library-file window. **No retained result in this project was generated under an actual race condition — but the 9 early-era and 3 middle-era scripts remain structurally vulnerable should they ever be rerun while a patch is in flight, and should be upgraded to the fully-hardened pattern before any future reuse,** which we flag as a concrete, actionable item rather than a resolved one. This also confirms directly (rather than merely inferring from file-edit timing, as an earlier version of this section did) that Sections 7.9’s and 7.10’s results do not need regeneration on this account. As disclosed at the top of Section 7, **essentially the entirety of Sections 7.1–7.8 predates verified-clean generation** and should be read with that caveat; critically, Section 7.10.3’s frozen confirmation set — previously stated as “generated entirely post-fix” — has since been shown to be an internally mixed pre-/post-patch artifact, not uniformly post-fix as this document previously claimed. That claim is withdrawn. A genuinely verified-clean regeneration of the frozen confirmation set has since been produced under the fully-hardened pipeline; see the reproducibility appendix for its status.

7.10.6 Signed-volume sign convention — root cause found, exact formula and fix now supplied. The “signed executed volume” feature used in Study 4 (Section 7.2) is computed as `buy_vol - sell_vol` over executed trades in a time bin, where `buy_vol/sell_vol` are summed by the `side` field ABIDES’s order book logs for each execution. **The in-**

version’s exact mechanism, traced to source: ABIDES’s order book logs the *passive/resting* order’s side at execution, not the aggressor’s — `side="SELL" if order.side.is_bid() else "BUY"`, where `order` is the incoming aggressor order, so an aggressor buy (hitting a resting ask) is logged as `side="SELL"` (the resting order’s side), and vice versa. The feature’s code therefore computes (aggressor-sell-initiated volume) — (aggressor-buy-initiated volume), the exact negation of what the variable name and this paper’s documentation describe as “buy-initiated minus sell-initiated” pressure. **The fix is a single negation** — `signed_exec_vol_CORRECT = sell-labeled qty - buy-labeled qty` in the current logging convention, equivalently, flip the sign of every value the current code produces. This does not require access to any information a real market participant couldn’t have: the passive/aggressor distinction is recoverable from ordinary trade-and-quote data (compare execution price to the prevailing best bid/ask at execution time), so the feature remains operationally valid once correctly signed — this was a pure implementation bug, not a hidden-information problem. As previously established, this sign error had zero effect on any reported accuracy or AUC (a linear classifier learns the data’s true orientation regardless of which label the documentation assigns it), so no classification result in this paper needs re-verification; only the Methods description of this feature and its worked-example direction need updating to match the corrected formula above.

7.10.7 Severity extrapolation — renamed per external review, since this tests magnitude generalization within the same mechanisms, not held-out mechanisms. An earlier version of this section’s title (“held-out mechanism generalization”) overstated what was tested: the table below varies incident *magnitude* outside a training range within the *same* four constructions (three independent designs plus one feature variant, not four independent designs), not generalization to a mechanism never seen during training or calibration. A genuine held-out-*mechanism* test (training on three designs, evaluating on a fourth never seen during training) is reported separately below in this same subsection.

Design / feature set	In-range (held-out)	Out-of-range (extrapolated magnitude)	Pattern
Original, 4-channel	0.600 acc / 0.518 AUC	0.780 / 0.800	Improves (expected: higher severity is an easier signal, not evidence of memorization)
Original, Study 4 expanded features	1.000 / 1.000	1.000 / 1.000	Saturated
Study 5	0.867 / 0.821	0.540 / 0.558	Accuracy drops substantially out-of-range
Third design	1.000 / 1.000	0.960 / 0.998	Negligible drop

Three of four rows generalize cleanly or improve out-of-range, the expected pattern when the held-out range is more severe, not evidence of memorization. **Study 5 moves in the opposite direction, and we state two candidate explanations without claiming to have established which is correct, per external review:** (a) Study 5’s in-range estimate (0.867, $n = 15$) was an optimistic small-sample draw, and the larger out-of-range test ($n = 50$, 0.540) is closer to Study 5’s independently-established near-chance baseline (Section 7.2’s Study 5 result: accuracy CI [0.446, 0.608]); or (b) this reflects a genuine distribution shift specific to this design. We do not have matched sample sizes between the in-range and out-of-range tests, which would be needed to distinguish these two explanations, and we withdraw an earlier claim that (a) is “the most likely explanation” — both remain open pending a dedicated follow-up with matched sample sizes.

A genuine held-out-mechanism test, distinct from the severity-extrapolation table above — renamed leave-one-design-out generalization per external review, to make clear that compromise and fault remain the two known cause labels throughout: training on three of the four independent designs and evaluating on the fourth, held out entirely from training and calibration. Results were mixed, and we correct our own initial interpretation of the below-chance AUCs, which understated

what they actually show. **A below-0.5 AUC is not “no discriminatory structure” — for a real-valued score, $\text{AUC}(-s) = 1 - \text{AUC}(s)$, so an AUC of 0.137 corresponds to a *reversed* ranking with an effective discriminative AUC of 0.863 under the opposite orientation, and 0.084 corresponds to 0.916.** These are strongly reversed rankings, not an absence of structure, and are compatible with the held-out design’s score orientation failing to transfer from the three training designs, rather than the classifier learning nothing transferable. (To be explicit about what we are not claiming: reversing a score after inspecting held-out labels would not be a valid way to report a deployment improvement — the complement is a diagnostic interpretation of what happened, not a proposed correction to apply going forward.) In the separated regime: the original design generalized moderately when held out (accuracy 0.609, 95% CI [0.557, 0.660]; AUC 0.726, CI [0.673, 0.775]; $n_{\text{train}} = 900$, $n_{\text{test}} = 350$); the third design’s held-out AUC was 0.137 (CI [0.097, 0.182] — reversed-ranking, per above, and notably an interval **entirely below 0.5**, ruling out chance as an explanation for this specific reversal; $n_{\text{train}} = 950$, $n_{\text{test}} = 300$); Study 5 showed chance-level generalization (accuracy 0.460, CI [0.403, 0.513]; AUC 0.470, CI [0.405, 0.534] — this interval does straddle 0.5); the fourth design showed weak-to-moderate generalization (accuracy 0.583, CI [0.530, 0.640]; AUC 0.603, CI [0.539, 0.668]). In the confounded regime: held-out AUCs and their 95% CIs were 0.369 [0.310, 0.430] (original), 0.381 [0.316, 0.441] (Study 5), 0.388 [0.324, 0.450] (fourth design) — all three straddling or bordering 0.5, consistent with weak-to-chance generalization — except the third design’s 0.084 [0.055, 0.120], again a reversed ranking with an interval entirely below 0.5, not attributable to chance. **All 8 cells’ point estimates, confidence intervals (2,000-resample bootstrap of the held-out test set), and exact train/test counts are provided in full as supplementary data (table_leave_one_design_out_full.csv), closing the gap flagged by external review that this table’s full evidence was described but not supplied. We read this, at the corrected strength, as evidence that classifier score orientation is substantially construction-specific and does not reliably transfer across independently-designed mechanisms — sometimes failing to transfer at all, and at least twice transferring in reversed form — which is a more specific and more interesting finding than “no transferable signal,” and still a genuine limit on generalizing any single construction’s classifier to a different one.**

Provenance caveat. The leave-one-design-out results in this subsection were computed on the original feature files and have not been re-run on the regenerated (Era-3, large-N) data used everywhere else in this paper as of the unification round. For the original and fourth designs this is immaterial (their regenerated features are bit-identical); for Study 5, the third design, and Study 4, whose regenerated data changed (Section 7.10.9), the held-out cells should be re-run before submission or this subsection explicitly scoped as a pre-regeneration analysis.

The root cause of the third design’s reversal has since been identified directly, closing what was previously listed as required follow-up. The pipeline itself is clean: the classifier (`LogisticRegression`), feature set (the identical 12-dimension spread/depth/cancel-rate/mid-price set across all four designs, confirmed by matching array shapes), and train/test split are confirmed identical across all four held-out configurations, with the held-out design’s data loaded from a separate file never passed to `.fit()` — no leakage or pipeline inconsistency. The reversal is **not** a signed-volume sign-convention issue, since signed volume is not among the 12 features used in this specific experiment. Examining the pooled classifier’s learned coefficients (trained on the other three designs) against the third design’s own class-conditional feature means: 6 of 12 features show a $K-F$ direction in the third design opposite to what the pooled classifier learned to associate with K — not a majority of features, so the reversal is not simply “every feature flipped,” but concentrated in a smaller number of high-coefficient-magnitude features, particularly `spread_mean` (the pooled classifier’s largest-magnitude coefficient). **The mechanistic explanation:** the third design’s incident agent has no directional (`buy_bias`) dimension at all — side is an unbiased coin flip for both K and F — so its actual K -vs- F distinction is driven entirely by size, rate, and cancellation-rate differences, a structurally different axis from the other three designs’ predominantly directional distinction. A classifier trained mostly on directional signal, applied to a design whose real distinguishing signal lies elsewhere, does not merely fail to generalize — it applies a decision rule that happens to be anti-correlated with the third design’s actual (non-directional) mechanism, producing the observed reversed ranking rather than a null result.

7.10.8 Error rates alongside cost — the decomposition holds exactly, under the correct (positive-fault-loss) identity. Reporting false-positive and false-negative rates alongside accuracy/AUC for every cost table confirms the identity $R(\pi) - R_{\text{oracle}} = p(H_K - C_K) \cdot \text{FNR} + (1 - p)(C_F -$

H_F) · FPR exactly (verified to within floating-point precision) — **this replaces an earlier, stale statement of the same identity that omitted the $-H_F$ term, appropriate only for Section 6’s own zero-fault-loss toy scenario, not Section 7’s positive-fault-loss model** — and shows that threshold sensitivity — how much a policy’s cost depends on exactly where the decision threshold is set — is concentrated almost entirely in the weaker classifiers (Study 5 and the original contrast construction). **We correct an overclaim here, per external review: Study 4 and the third design’s separated-regime error rates are indeed near zero, but their *confounded*-regime FNRs are not** (Study 4: 0.305; third design: 0.179, per Table 4) — “already near zero on both sides” was accurate only for the separated regime and is corrected to that scope.

A terminology blur corrected here, per external review: two different thresholds appear in this paper and should not be conflated. $q^* = (C_F - H_F)/[(H_K - C_K) + (C_F - H_F)]$ (Section 6) is fixed *on the deployment posterior* — it depends only on the loss table, not on the prior p , and is the threshold applied to a correctly prior-adjusted score. A separate quantity, $s^*(p)$, is the *equivalent* cutoff applied directly to a raw score calibrated to a different (e.g., balanced 50/50) training prior — this one *does* vary with p , since it must absorb the prior-shift correction that q^* does not need to. Where this section and Section 6 refer to “the cost-optimal threshold” without further qualification, this means q^* applied to the prior-adjusted score, not $s^*(p)$ applied to the raw one — the two coincide only when p equals the training prior exactly.

A further caution, per external review: mean FPR and mean FNR across five priors, multiplied by a mean prior, do not reproduce mean cost — the cost-optimal threshold itself changes with the prior, so per-prior rates and per-prior weights are associated, not independent; the per-prior decomposition (not shown in full here) should be checked before aggregating, and the by-design-and-regime means reported in Section 7.7 are simple averages of the per-prior figures, not a re-derivation from averaged rates.

7.10.9 Final end-to-end regeneration under the hardened pipeline — closing the last provenance gap, and finding two real changes in the process. The run-manifest audit (Appendix: Reproducibility) established that Table 4’s *classifier training data* — as distinct from the cost pipeline built on top of it, which was already verified — predated the RNG-audit patches for every profile, and had never been regenerated under the fully-hardened (spawn + maxtasksperchild=1 + runtime fingerprint verifi-

cation) pipeline for Study 5, the third design, or Study 4. **All three have now been regenerated end-to-end at large N (1,000 episodes per class), closing that gap; every profile in this paper now has at least one large- N hardened-pipeline regeneration.** Two of the six regenerated cells changed meaningfully, and we report both rather than only the reassuring one:

Design / regime	Pre-patch acc/AUC	Hardened, $N=1000$ /class	Change
Study 5, separated	0.522 / 0.588	0.615 / 0.661	+9.3pp acc, +7.3pp AUC
Study 5, confounded	0.544 / 0.547	0.558 / 0.579	+3.2pp AUC
Third design, separated	0.989 / 1.000	0.985 / 0.999	No meaningful change
Third design, confounded	0.889 / 0.969	0.845 / 0.911	−4.4pp acc, −5.9pp AUC
Study 4, separated	0.991 / 1.000	0.995 / 1.000	No meaningful change
Study 4, confounded	0.886 / 0.933	0.870 / 0.938	No meaningful change

Two findings worth stating explicitly. First, **the third design’s near-perfect separated-regime result is confirmed genuine** — it survived the patch essentially unchanged, ruling out the RNG bug as its explanation, which is exactly the kind of targeted test that distinguishes a real result from an artifact. Second, **the same design’s confounded-regime result did not survive as well:** its apparent near-separability was partly inflated by the unpatched RNG, and the corrected figure is meaningfully lower. Study 5, meanwhile, moved in the opposite direction — its classifier is *more* discriminative than the pre-patch data suggested, though still the weakest of the five profiles. **The direction of these corrections is mixed, not uniformly favorable, which is what we would expect from a genuine bug fix rather than a selective re-analysis.** Propagating the three meaningfully-changed cells through the cost pipeline shifts $\hat{R}_p$ by between roughly \$0 and \$47K (the largest at the third design’s confounded regime, $P(K) = 0.50$), again directionally mixed rather than uni-

formly worse or better. One production code change was made in the course of this work: `config_builder_v3.py`, which the manifest audit found had never been patched for bugs A/B at any point in the project, was permanently patched.

7.11. Consequence for Paper 2 — a research question raised, not a settled counter-recommendation

Paper 2’s Corollary 2 argues that verification investment should be targeted specifically at the belief range where this paper’s confounding condition holds, since that is exactly where passive observation is powerless. **A terminology precision, per external review, worth stating before anything else: “the confounded belief range” conflates two genuinely separate things.** Whether K and F are observationally confounded (Theorem 1’s condition, $\theta_K C b_K = \theta_F C b_F$) is a property of the fixed mechanism instances and the observation law — it does not depend on the prior p at all. What *does* depend on p is the *value* of resolving that confounding: Proposition 1’s optimal ambiguity cost, $\min\{(1 - p)\text{DISRUPT}, p(\text{HARM} - C_{OP})\}$, varies with p and is zero exactly when $p \in \{0, 1\}$. Paper 2’s Corollary 2, read precisely, is a claim about *this second, prior-dependent quantity* — verification is worth its cost over some range of p given a fixed confounding instance — not a claim that confounding itself “holds” only for certain priors. This paper’s findings raise real questions about that recommendation’s premises, and we state them as open questions requiring further work, not as an established counter-recommendation — Paper 2 is a companion result we have not reviewed in this document, and narrowing or restating its conclusions would require a specified value-of-information model (verification’s own accuracy, cost, and timing, compared explicitly against the passive alternative), which this paper does not provide.

First: Section 4.1’s Theorem 2 shows that even *within the linear model Corollary 2 itself is stated for*, the confounded region is not a stable target — it survives only under the special (measure-zero) condition that the *specific incident difference* Δ_0 lies in the unobservable subspace of the dynamics (not that the dynamics preserve the *entire* kernel of C — a stronger and unnecessary condition we state precisely here per external review, since a single vector can lie in an unobservable subspace without that subspace containing all of $\ker(C)$), which fails generically once the incident is modeled as a persistent behavioral change rather than a one-off sensor blip. **Second,** this section’s opening reframing means Corollary 2’s premise (that the confounded belief

range is even well-defined for a real incident) was never established in the first place — the ABIDES constructions only ever matched one summary statistic, not the full observation law Theorem 1 and Corollary 2 require. **Third**, the sensitivity analysis sharpens the concern further: which real incidents produce which classification profile is highly construction-dependent — four independently-designed constructions (plus one feature-engineering variant of the first) produced five different classification profiles between them, not five independent constructions (a terminology distinction worth being precise about, since Study 4 shares its underlying construction with the original design and differs only in feature access). **Fourth**: Section 7.10.4’s re-verified cost analysis shows a genuinely mixed picture — one previously-flagged profile (Study 5) shows a small positive fitted-classifier regret relative to the best constant policy once correctly calibrated (a weaker, more precisely-scoped claim than “carries no exploitable cost information,” which an earlier version of this sentence overstated — we have observed little measured benefit from this specific fitted policy, not established that the underlying observation law is exactly confounded or that no policy could extract value), but another (the original contrast construction) turns out to be a case where a passive classifier provides real, if modest, value once correctly calibrated. **None of these four points, individually or together, establish that verification is unnecessary where a classifier looks useful, or that it is worth its cost where a classifier looks weak** — a useful classifier does not by itself resolve whether verification would have been better, and neither the belief threshold at which Corollary 2’s condition holds, nor any actual verification channel’s accuracy or cost, is defined or evaluated anywhere in this paper. **What we can defensibly say is narrower**: an operator or researcher applying Paper 2’s recommendation should not assume the confounded belief range is fixed, well-defined, or easy to identify from construction labels alone, given the instability documented in Sections 4.1 and 7 — this is a caution about Corollary 2’s premises, not a demonstration that its practical recommendation is wrong. Resolving this properly requires Paper 2’s own update, incorporating Theorem 2 and this section’s findings, and an explicit value-of-information comparison neither paper currently provides.

8. Discussion

The identifiability boundary from the linear model is not worthless — Sections 3–6 show it is exact and reproducible within its stated scope, and it remains a legitimate diagnostic wherever that scope holds (plausibly, small-perturbation regimes around a steady-state book). **But Section 4.1 already shows that scope is narrower than Theorem 1 alone suggests:** confounding that survives a one-shot observation check is not confounding that survives feedback into realistic dynamics, and the gap between the two is governed by a well-understood classical quantity (the unobservable subspace of (A, C)) rather than anything exotic. This gives a first, purely linear-model reason to distrust a naive reading of Theorem 1 as operationally robust, before nonlinearity is even introduced. Section 7’s empirical study of ABIDES adds a second, independent caution, at the correct strength: the study was never a test of Theorem 1, since none of the tested constructions were shown to instantiate the theorem’s actual premise (full trajectory-distribution equality) — what it shows is that attempting to construct such an instantiation in a realistic multi-agent market is itself difficult and sensitive to construction choices that are, per a direct audit (Section 7.3), mostly representable-but-unexplored on the linear-model side rather than genuinely structurally absent from it. **We narrow an earlier, overstated version of this claim: calibration-by-search is not a general structural dividing line between linear and nonlinear systems — the defensible statement is that this specific, known-parameter linear construction has an analytic matching solution, while this specific ABIDES construction required numerical search, a fact about these two particular constructions, not a general property of linearity versus nonlinearity.** A third caution, arising from this project’s own development process rather than the theory itself (detailed in the supplementary changelog): a decision-cost claim is only as trustworthy as the policy used to generate it, and a plausible-looking, sufficiently surprising empirical result should prompt checking the decision rule before checking the theory. The broader methodological point across all three cautions, applicable beyond this paper: **an identifiability boundary proved for a linearized model is a claim about that model, not a claim about deployment, until the transfer is actually established**, and neither the theoretical fragility (Section 4.1) nor the empirical construction-dependence (Section 7) nor a cost-analysis finding should be taken on the strength of a

single striking result without independent verification.

9. Limitations and what remains before submission

We are explicit that this is a **complete draft with a genuine theoretical core, a working pilot-scale validation, and a working nonlinear validation** — not yet a submission-ready empirical study at the standard a Q1 venue will expect. Specifically, before this manuscript is submission-ready:

1. ~~Scale up the simulator~~ **Done**. The incident-generation layer has been wired into ABIDES, a real agent-based market simulator, with realistic order-book mechanics and a 95-agent background population (Section 7). Remaining: the full scenario-family taxonomy (behavior, regime, and combinations) beyond the two mechanisms (compromise, fault) treated here.
2. ~~Validate the linearization~~ **Corrected, not done — this item’s earlier “Done” status is retracted**. Section 7’s opening is explicit that no linearization bridge (operating point, Jacobian, or approximation-error bound connecting the linear model to ABIDES’s dynamics) has been derived, and that this section is an independent empirical comparison, not a validation of the linear model by the nonlinear one. What *is* done, and should not be confused with linearization validation: a five-part sensitivity analysis of how ABIDES classification/cost behavior responds to feature, footprint, and classifier choices, now backed by bootstrap confidence intervals on every load-bearing claim (Section 7). The threshold-rule bug (Study 2) has been fixed, not just flagged — the corrected result is a genuine, mild dissent from the reversal, not noise. Studies 4 and 5 now carry the same statistical confidence as Study 1’s original reversal, and Study 5’s initial “no signal” framing has been corrected to a more precise (and more interesting) partial-signal finding. Remaining: real cost calibration (item 3, unchanged), confirming the practitioner checklist (Section 7.4) against feedback from external review (item 8, below), and — if a linearization claim is ever wanted — actually deriving the bridge this item previously and incorrectly claimed was complete.

3. **Realistic cost calibration.** Unchanged from the prior draft: the sensitivity sweep (Section 6.1) shows the qualitative *linear-model* finding is not an artifact of the specific cost triple, but real market-impact calibration is still needed, and now additionally needs to be re-run on the ABIDES-derived costs given Section 7’s reversal.
4. ~~Baseline breadth~~ **Partially done, and this item’s earlier “has not yet been repeated” claim is stale and corrected here.** The chance-level result under linear-model confounding is confirmed across four classifier families (Section 5.1). On the ABIDES side, Section 7.2’s Study 2 also tested four classifier families (logistic regression, Gaussian naive Bayes, random forest, and a threshold rule) on the original construction specifically — this has been done, contrary to what this item previously claimed. What remains genuinely unrepeated: the same four-classifier sweep has not been run on the other three ABIDES designs (Study 5, third design, fourth design), so classifier-choice sensitivity is confirmed for one construction, not yet checked across all four.
5. ~~Second, independent implementation~~ **Done for the linear model** (Section 6, and Section 7’s cross-check). **For ABIDES, now explicitly classified rather than left as an open aspiration:** an independence audit of every retained result found that all four injection designs (original, Study 5, third, fourth) share the same underlying `config_builder`/kernel/feature/cost-pipeline code — they are independently-*parameterized* variants, not independently-*coded* implementations. Zero results in this paper’s ABIDES component qualify as a genuine second implementation in the sense Section 6’s linear-model cross-check does; a bug shared across all four designs’ common infrastructure would not be caught by any comparison currently in this paper. A true second implementation — a separately-coded ABIDES harness, not merely a different calibration target within the existing one — remains a real gap, not a nuance of the existing one.
6. **Feature and population sensitivity, flagged by the companion validation report.** The ABIDES result depends on a fixed, hand-chosen feature set and a single population configuration; the companion validation report (not reproduced in this manuscript) notes that mid-price had to be added to the feature set after its absence made the separated case undetectable by construction. **We flag, rather than resolve, that this specific episode is not otherwise narrated anywhere in Section 7’s own text** — Section 7.5’s feature-

sensitivity story centers on signed executed volume, not mid-price, and we have not reconciled the two accounts within this manuscript. Either the validation report’s claim should be incorporated into Section 7 with its own supporting detail, or the discrepancy between the two documents should be resolved, before this item is treated as settled.

7. ~~Missing citations~~ **Substantially done.** The bibliography now runs to 26 entries (numbering 1–24, including 11b and 21b) — earlier versions of this item said 22, then 25, both undercounts; corrected here. **A stale self-contradiction, corrected here per external review:** Shu et al. (2021) was in fact located and fully cited (entry 16, with confirmed title, venue, and DOI) several rounds ago, but two separate notes in this document continued to say it “could not be independently located” — a leftover from before it was found. The remaining open question about this entry is not its existence but its *relevance*: whether this CV/domain-generalization paper genuinely bears on the trading-incident-identification setting, or was miscited in the original thesis proposal, is still unresolved and needs the original proposal author’s input.
8. **External review**, now with an additional, specific ask: given Section 7’s reversal, review should explicitly include someone who can assess whether the ABIDES calibration and feature choices are representative of real gateway telemetry, not only someone from the distinguishability-theory or agent-based-simulation communities separately.
9. **Disentangling Theorem 2’s state-feedback fragility from Section 7’s empirical factors — attempted (Section 7.8), not cleanly resolved.** A persistence ablation found a large effect directionally consistent with Theorem 2’s prediction (removing persistence collapsed classifiability from 0.829 to 0.478 accuracy, near-exact chance, in the summary-matched case), but external review correctly identified that this design conflates persistence with incident duration and cumulative activity, so it does not isolate the mechanism cleanly. A follow-up reactive-vs-replayed redesign (Section 7.10.2) was built specifically to isolate this but is inconclusive at the sample size tested ($N = 30/\text{cell}$) — this remains genuinely open.
10. **A fourth injection design (Section 7.9) found a fifth distinct outcome, exactly as Section 7.6 predicted it might** — this is not a new gap, but it does mean the “space of possible outcomes” remains open-ended rather than closed by any fixed number of designs;

a future version of this work could pursue the systematic characterization Section 7.6 already flags as future work (a structured sweep over attribute-subset/calibration-target combinations) rather than continuing to add designs one at a time.

11. ~~The corrected cost pipeline needs re-verification~~ **Done, and fully recomputed (Section 7.10.4/7.7)**, combining both the prior-shift threshold fix and the positive-fault-loss fix in one pass rather than reporting them separately at inconsistent dollar scales. Study 5 remains the one profile with no measurable classifier benefit over the best constant policy; the other three designs (including V4, now included) substantially beat the best constant. This is now a resolved, re-verified item at the correct dollar scale, not an open question.
12. **Nearly all ABIDES results in Sections 7.1–7.8 predate a since-completed RNG fix** (Section 7.10.5) and should be regenerated under the fixed RNG before this paper is treated as submission-ready. **We no longer cite the frozen confirmation set (Section 7.10.3) as reassuring evidence here:** a deeper investigation (Section 7’s opening disclosure, and the Appendix) found that claim of full post-fix provenance was itself incorrect, due to a worker-process import-caching bug, and it has been withdrawn — regeneration under a genuinely verified-clean pipeline, not reliance on that set, is the actual outstanding requirement.

None of these are small. What this draft now establishes is that the theoretical core is sound and provable (Sections 3–6), *and* that its practical relevance is genuinely uncertain and possibly narrower than assumed (Section 7) — which is itself a publishable, non-obvious finding, not merely an unfinished validation.

10. Conclusion

We gave a linear-Gaussian model of gateway-observable order flow and proved an exact fixed-instance equality criterion — $\theta_K C b_K = \theta_F C b_F$, together with a separate rank-based characterization of which severity/direction pairs *can be made* to satisfy it — for when a cyber compromise and a benign fault are observationally indistinguishable from an operator’s passively

available evidence (the Introduction and this Conclusion previously both called this a single “rank-based necessary and sufficient condition,” which conflates the fixed-instance equality with the separate existence question; corrected here to match the Abstract’s more precise statement), together with a proof that this imposes an unavoidable cost floor — equal to the best-achievable-using-observations risk minus the oracle’s risk — on any policy relying on that observation alone. This part of the result is exact, and an independently implemented second simulator confirms it is not a code artifact; the optimal ambiguity cost is, under an illustrative (not empirically calibrated) Almgren–Chriss-based dollar reconstruction, approximately \$41,000 per incident-episode at the illustrative scenario’s assumed base rate. We then showed, without leaving the linear model, that this confounding is itself far more fragile than Theorem 1 alone suggests: modeling the incident as a state-level rather than output-level disturbance requires confounding to lie in the classical (Kalman) unobservable subspace of the dynamics — generically the trivial subspace, though the specific pair-relevant subspace, not the entire kernel of C , is what actually matters (see Section 4.1’s corrected statement). We then conducted a controlled empirical study — explicitly not a test of Theorem 1, whose premise (full observation-trajectory-law equality) none of our simulated constructions were confirmed to satisfy — of how incident-construction, feature-access, and calibration choices affect classification and decision cost in ABIDES, a realistic agent-based market simulator. Across four independently-designed injection constructions, several qualitatively distinct classification profiles emerged, sensitive to choices that a direct audit (Section 7.3) found to be mostly representable-but-unexplored on the linear-model side rather than genuinely absent from it. **We narrow an overstated claim here:** an earlier version of this paragraph called calibration-by-search “the one factor genuinely absent” from the linear model — the more defensible statement is narrower still: *this specific, known-parameter linear construction* has an analytic matching solution via C ’s nullspace, while *this specific ABIDES construction* required numerical search; this is a fact about the two particular constructions used, not a general structural claim that linear systems never need search or that nonlinear systems always do. We conclude that the linear model’s identifiability and cost-floor results are exact and reproducible within the regime they are proved for; that this regime is narrower than a first reading suggests, both because of Theorem 2’s state-feedback fragility and because engineering a valid nonlinear analog of exact confounding turns out to be a genuinely difficult, easy-to-get-wrong task

in the specific constructions attempted here, not a demonstrated general property; and that a decision-cost claim built on a misspecified policy can manufacture findings in either direction — sometimes inflating harm from nothing, sometimes hiding real benefit, a methodological lesson this project learned firsthand during its own development and reports in detail in the supplementary changelog rather than the main text. The paper’s defensible contribution is a corrected theoretical core plus a controlled, honestly-scoped empirical study of observation-matching and decision cost in a simulated market — not a demonstrated failure of linear theory under nonlinear conditions, and not a general claim about calibration difficulty beyond the specific constructions tested. This also means the companion paper’s (Paper 2) original recommendation to target verification investment at a theoretically-defined confounded region cannot stand as first stated, since that region’s practical identity was never established as instantiating either paper’s premise for a real incident, and the cost consequences of misclassifying it depend on calibration details neither paper originally accounted for. The companion paper is being restated accordingly: verification is treated there as a costed action on a separate information channel (authorization, not order flow — a channel Theorem 1’s observation law does not cover), and its value is a prior-dependent quantity derived from Proposition 1’s cost floor, which yields a two-threshold leave/verify/contain rule rather than a region defined by the confounding condition alone.

Appendix A. Reproducibility

A concurrency defect in ABIDES log writing, found and fixed during the unification round. ABIDES’s `Kernel.write_summary_log` names its output directory by wall-clock timestamp at one-second resolution; parallel workers finishing in the same second collide on `./log/<timestamp>/` (a time-of-check/time-of-use race). The failure is loud — the affected episode crashes — and never corrupts data silently, so no retained result is affected. The fix, applied to all runs from the unification round on, gives each worker its own working directory.

Exact train/test denominators for every “perfect accuracy” result in this paper, per external review’s request — needed to verify

each Clopper-Pearson interval independently: V4 confounded (original small-N run): 300 total / 210 train / 90 test, interval [0.9598, 1.0]; V4 confounded (large-N verified-clean run, the one cited in Section 7.9): 2,000 total / 1,400 train / 600 test, interval [0.9939, 1.0]; Study 4 contrast (expanded features): 350 total / 245 train / 105 test, 1 error observed (accuracy 0.9905, not exactly 1.0), interval [0.9481, 0.9998]; third design (V3) contrast: 300 total / 210 train / 90 test, interval [0.9598, 1.0].

Sections 5–6 results were generated by two self-contained Python scripts (`sim.py`, `cost_quantification.py`, plus `cost_sensitivity.py` and `baseline_robustness.py` for the robustness checks in Sections 5.1/6.1, each subsequently re-run across 5 seeds with bootstrap confidence intervals), using NumPy and scikit-learn, fixed random seeds. **Section 4.1’s Theorem 2 was validated by two further scripts** (`state_vs_output_injection.py` and `coupled_A_test.py`), the first of which surfaced and explained a null result (the paper’s own $A = 0.6I$ trivially preserves confounding under feedback) before the second, using a genuinely coupled A , confirmed the predicted fragility — both scripts and their output are included for transparency about that two-step process rather than presenting only the confirming result. Section 7’s ABIDES results were generated by an incident-injection layer built on a local clone of ABIDES (`abides-core`, `abides-markets`), full harness (`incident_agent.py`, `config_builder.py`, `features.py`, `regimes.py`, `calibrate.py`, `run_episode.py`, `batch_run.py`, `task3_classification.py`, `task4_cost_quantification.py`), an independent second linear-model implementation (`independent_linear_sim.py`), three further independently-designed ABIDES injection constructions beyond the original (Study 5, the third design of Section 7.5, and the fourth design of Section 7.9 — **four injection layers in total including the original; an earlier version of this sentence said “three... including the original,” omitting the fourth design entirely, corrected here**), a footprint sweep, a per-feature ablation, and an Almgren–Chriss illustrative dollar-scale reconstruction (Section 6.2; not a “calibration” — this word is avoided per Section 6.2’s own correction, and this occurrence is fixed to match), all detailed in the companion validation report. **Supplementary data, added per external review’s F6 finding that the full evidence tables were described but not supplied:** `table4_full_50cell.csv` (all 50 design $\times$ regime $\times$ prior cells underlying Section 7.7’s summary, with classifier/constant/oracle costs, FPR/FNR, threshold, sample counts, and both “beats always-contain”/“beats best-constant” booleans with the stated criterion, now using the direct class-conditional weighting method)

and `table_leave_one_design_out_full.csv` (all 8 held-out-design cells underlying Section 7.10.7, now including exact train/test counts and 95% bootstrap confidence intervals for both accuracy and AUC).

A run-manifest audit, per external review’s F1 finding that no explicit mapping from published result to actual run existed, was built and surfaced a genuine provenance gap — which has since been closed (Section 7.10.9). This project’s ABIDES fork is based on commit `f9cbe51342b7dedd9587e4e069040d68a5c6477f` (2023-12-13), with four locally-modified files (three implementing the RNG-audit patches A/B/C, one an unrelated fill-price logging fix). **The classifier training feature data originally underlying all 50 cells of Table 4 — every one of the five profiles, both regimes — was generated between 2026-09-08 and 2026-09-09, before the RNG-audit patch existed at all, and in three of five cases (Study 5, the third design, the fourth design) before that specific design’s own config-builder file was individually patched, with the third design’s config-builder never patched at any point until this gap was addressed.** The cost-pipeline computation built on top of this data (Table 2c’s H_F values, the cost-optimal thresholds, Section 7.7’s direct class-conditional weighting) was verified under the fully-hardened pipeline throughout — but the classifier predictions feeding into that pipeline were not, until now.

This gap is now closed: all five profiles have been regenerated end-to-end at large N under the fully-hardened pipeline (Section 7.10.9), and `config_builder_v3.py` has been permanently patched. The regeneration was not a formality — two of six regenerated cells changed meaningfully, in opposite directions (Study 5’s separated regime improved by 9.3pp accuracy; the third design’s confounded regime fell by 4.4pp accuracy, revealing that part of its apparent near-separability had been an RNG artifact). Section 7.10.9 reports the full before/after comparison and the resulting cost-pipeline shifts. We note that this outcome retroactively justifies treating the gap as a real risk rather than a formality: the indirect reassurance previously available (V1 and V4’s large- N reruns reproducing their originals closely) would *not* have predicted the third design’s confounded-regime change, and assuming the gap was clean by extension from those two designs would have left a real error in the paper.

Two reproducibility notes worth stating in the paper itself, not just the report (with, per a later completeness audit, the previously-missing base commit now on record: this project’s ABIDES fork is based on `jpmorganchase/abides-jpmc-public`

commit `f9cbe513...` dated 2023-12-13, with four locally-modified files beyond the `str_to_ns` and RNG patches described below — a detail that should have been in this appendix from the start and is a genuine reproducibility gap now closed): (1) we found and fixed a real bug in ABIDES’s `str_to_ns()` utility under pandas $\geq 2.x$ that silently mis-scaled every simulated time interval by 1000x with no crash or warning; any prior result produced with this codebase under a modern pandas version should be treated as suspect until checked against this fix. (2) We separately found a bug in ABIDES’s `get_wake_time()` utility that draws from an unseeded global random number generator, meaning identical seeds do not produce bit-identical episodes across separate process invocations (only within one) — this bug, along with two further unseeded-RNG bugs found later (Section 7.10.5), has since been patched, and bit-identical cross-process reproducibility was demonstrated after all three fixes were applied. **A fourth, subtler bug was then found in the verification process itself:** forked worker processes cache their first lazy import of the patched modules for their entire lifetime, so a result generated after the on-disk patch could still reflect pre-patch code depending on exactly when the worker that produced it first imported the module — this affected several results previously believed to be clean, including the frozen confirmation set, which this document previously and incorrectly described as “generated entirely post-fix.” That claim is withdrawn; a genuinely verified-clean pipeline (forced fresh re-import per task, plus a behavioral fingerprint check independent of file timestamps) has since been built and used to regenerate the affected results. Critically, **almost every ABIDES result in this paper (Sections 7.1–7.8) was generated before verified-clean generation was possible**, and should be understood as statistically valid but not bit-for-bit reproducible from the stated seeds, and in a small number of cases (now identified and being regenerated) may have been generated under an unverified mix of pre- and post-patch code despite appearing to postdate the fix by file timestamp alone. This distinction — and the fact that most of the paper’s numbers predate verified-clean generation — should be disclosed plainly in any submission using this codebase, not treated as a minor footnote.

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